
JOURNAL OF INQUIRY-BASED LEARNING IN MATHEMATICS
No. **42**, (May 2016)

Introduction to Advanced Mathematics

B. J. Baker
LaTeXed by Alys Rodriguez

Lamar University

Contents

To the Instructor	v
To the Student	viii
1 INTRODUCTION TO LOGIC	1
1.1 STATEMENTS	1
1.2 DENIALS	3
1.3 TRUTH TABLES	3
1.4 COMPOUND STATEMENTS JOINED BY AND/OR	3
1.5 LOGICALLY EQUIVALENT STATEMENTS, TAUTOLOGIES, & CONTRADICTIONS	4
1.6 DENIALS OF COMPOUND STATEMENTS	5
1.7 CONDITIONAL STATEMENTS	7
1.7.1 DENIALS OF CONDITIONAL STATEMENTS . . .	7
1.7.2 UNIVERSAL AND EXISTENTIAL QUANTIFIERS	8
1.7.3 DENIALS OF STATEMENTS WITH UNIVERSAL AND EXISTENTIAL QUANTIFIERS	11
1.7.4 RELATED CONDITIONAL STATEMENTS	14
1.7.5 BICONDITIONAL STATEMENTS	15
2 METHODS OF PROOF	17
2.1 PRELIMINARIES	17
2.2 DIRECT PROOFS	18
2.3 PROOF BY THE CONTRAPOSITIVE	19
2.4 INDIRECT PROOFS	19
3 SETS AND SET OPERATIONS	22
3.1 SETS, ELEMENTS, AND NOTATION	22
3.2 AN AXIOM	23

3.3	DEFINITIONS	23
3.4	VENN DIAGRAMS	23
3.5	MORE DEFINITIONS WITH VENN DIAGRAMS	24
3.6	THEOREMS	26
3.7	DISPROVING A CONDITIONAL STATEMENT	28
3.8	CONJECTURES TO PROVE OR DISPROVE	28
3.9	VALID AND INVALID ARGUMENTS	29
4	RELATIONS AND EQUIVALENCE RELATIONS	31
4.1	DEFINITIONS DEALING WITH RELATIONS	31
4.2	EQUIVALENCE RELATIONS	32
4.3	THEOREMS ABOUT EQUIVALENCE RELATIONS	34
5	MAPPINGS AND BIJECTIONS	36
5.1	DEFINITIONS, NOTES, AND COMMENTS	36
5.2	DENIALS	38
5.3	A THEOREM ABOUT FUNCTIONS	41
5.4	BINARY OPERATIONS	41
6	THE SYSTEM $\mathbb{R}$ OF REAL NUMBERS	43
6.1	AXIOMS AND DEFINITIONS	43
6.2	THEOREMS	44
6.3	THE ORDER AXIOM AND MORE DEFINITIONS	45
6.4	ORDER THEOREMS	46
6.5	THE NATURAL NUMBERS AND THE LEAST NATURAL NUMBER AXIOM	48
6.6	THE FIRST PRINCIPLE OF MATHEMATICAL INDUCTION	49
6.7	THE EXTENDED FIRST PRINCIPLE OF MATHEMATICAL INDUCTION	52
6.8	THE SECOND PRINCIPLES OF MATHEMATICAL INDUCTION	53
6.9	RATIONAL NUMBERS	53
6.10	THE COMPLETENESS AXIOM	55
6.11	THEOREMS	57
	APPENDIX A	61

Notes to the Instructor**63**

To the Instructor

These notes were developed for a transition to higher mathematics course at Lamar University in Beaumont, Texas. Lamar University is a regional university with a current enrollment of approximately 14000. I teach this course as a highly modified Moore-method course, and I prefer to teach it on a 2-day a week schedule of 75-80 minutes each meeting. The best size for a class of this type is probably about 20 students. I have used these notes, with several minor (but continual) revisions, since 2006.

Topics include some logic, sets, equivalence relations, functions, binary operations, and the real number system, with each topic building on the previous topics. The real number system section also includes mathematical induction and the completeness axiom.

In the logic section explanations are relatively minimal and include a few examples; but in the rest of the material only definitions, notation, axioms, lists of theorems, and exercises are given. It is up to the instructor as to how much additional explanation and examples need to be given in any class.

When I teach this transition course, I try to cover the logic section as quickly as possible. Exercise 1 is gone over in class by calling on the students. In the other exercises, I use one problem as an example and assign the others for homework. Each problem is graded according to the number of items the student needs to get correct. For the truth tables, most of the problems are done as board work.

In the section on sets, the first 9 theorems are stated for completeness and are accepted without proof. However, I usually prove at least two of them (with as much input from the class as I can get), including one about the empty set (to get another example of an indirect proof), in order to illustrate how to use the tautologies from the previous section in writing valid proofs. Then of the remaining 10 theorems, I usually prove two of those in class (again with input from the class); then I select 2 of those theorems for the class to prove for homework. Each theorem is worth 10 points; a student

has 2 class meetings to get it turned in. If the proof is mostly correct, grades from 7-10 points, including half points, are given. Otherwise, an error is noted and questions asked and comments made to try to head the student in a correct direction. A notation of “try again” is made on the problem. If the student wants a grade on this theorem, he must try again to write a valid proof within 2 more class meetings; whereby he may get from 1 to 6.5 points, or 7 points if he does a really good proof. I do require that the proofs be correctly written and do make any corrections needed (even for relatively minor things but without taking off points). I try to teach the student to be as brief, clear, and precise as possible; and, here at the beginning, students are not allowed to skip steps.

Proofs of the remainder of the theorems are presented as board work by individual students. When a student is presenting a proof of a theorem at the board, it becomes “his” or “hers.” Other students are encouraged to ask questions, but are asked not to offer suggestions or otherwise help the student who is presenting. Again a proof presented at the board gets a grade of 7-10 points, including half-points, if mostly correct. If the proof has major errors, the student gets to try again until he/she succeeds. Working together on homework is allowed but not encouraged. The proof must be correctly written (with no skipped steps), and I do make corrections in wording.

The following is basically the manner in which I handle each section: 1) I go over the definitions and notes, giving examples as needed, 2) show how to work or prove a couple of problems or theorems, 3) assign a couple of theorems or problems for home work, and 4) leave the remainder for board work.

For a student’s grade, I give a midterm test (with a review sheet), a project (considered a take-home test consisting of 10 or 11 theorems and/or conjectures to prove or disprove), a second in-class test (also with a review sheet) at the end of the semester over material covered after the mid-term test, a comprehensive final exam, and a homework/classwork grade. The project is given in two parts, one part just after the midterm and the second part due just after the second test is returned. Students are given about two weeks to work on each part; they may use their own notes, home work, board work, and tests, but may not consult with any one else.

There actually is more material in the notes than can be covered in a semester unless you have really good students, but it is set up so that any section can be omitted or downplayed without causing problems. I try to cover the logic, sets, relations, and functions sections by the middle of the semester, and then the real number system the rest of the semester. I usually do not spend

much time on binary operations, the theorems after Axiom R-III about the natural numbers and integers, the Extended First Principle of Mathematical Induction, the Second Principles of Mathematical Induction, and the theorems after Theorem 50, and most of the time I do not get to Exercise 16. In those sections, I usually cover any definitions and/or theorems and state that we will just assume that the theorems listed are true. In general we do not prove all of the theorems in any section or work all the problems in any exercise. It just depends on what material you want to concentrate on and the length of time you want to spend on it.

At the level of this course, I do provide more hints and examples than I would in a higher level course, such as a course in analysis or topology. I consider a student at the level of this course like someone driving an old jalopy (that is, a car before the computer chip): you have to give him enough of a push to get him going so that he can then continue on by himself. He needs a basic understanding of what a proof is and what a counter-example is, i.e., some place to start. Few students I have taught know how to do that at the level of this course. The further a student progresses, the fewer hints and suggestions I give him or her.

Perhaps the best suggestion I can make to anyone who is contemplating using these notes is to try them out on one or two students in a guided independent study course.

To the Student

The object of this course is for you to learn the basics of proving theorems. There are three sayings which you may or may not have heard of, but which we suggest that you consider seriously with regards to this course.

The first one is “Mathematics is not a spectator sport.” You don’t learn mathematics by watching someone else do it. That’s particularly true when it comes to learning how to prove theorems. You need to learn how by proving theorems yourself, because it takes a different way of thinking than that which you are used to using in applying formulas to problems.

The second one is “The devil’s in the details.” The first and last steps in a proof are usually relatively easy. The hard part is figuring out all the in-between steps to take - the details.

The third one is “Stick with the nitty-gritty.” That means to stay with the basics. Make your statements as clear, concise, precise, and uncomplicated as possible. Try not to make the problem harder than it is. Mathematics can get complicated enough all by itself; you don’t need to help it. Learn the definitions (i.e., memorize them) and use them just about the way they are stated when applying them in a proof. The definitions are usually stated as precisely as possible. At the level of this course, you don’t want to skip steps. Most often, that’s where the errors occur: in the skipped steps.

These notes as we have written them were not intended to be used as a self study course. At the level of this course, the beginning level, you need an instructor, someone who understands how to do proofs, to give you feedback about what you have done, but not someone who just tells you the answers to your questions or just tells you what you should have done. You need to figure out as much as possible on your own. The material here presents you with ample opportunity to do just that.

Chapter 1

INTRODUCTION TO LOGIC

1.1 STATEMENTS

A **statement** is a declarative sentence which can be classified as true or false, but not both. The classification of “true” or “false” is called its truth value.

EXAMPLES: Classify each of the following sentences as a statement or a non-statement and explain why.

1. “The Great Salt Lake is in the state of Maine.” is _____
because _____.
2. “Argentina is a country in South America.” is _____
because _____.
3. “Where are you going?” is _____
because _____.
4. “Shut the door.” is _____
because _____.
5. “Look out!” is _____
because _____.
6. “ $x + 2 = 5$ ” is a declarative sentence, but since we don’t know what value x has, we can’t classify it as true or false because when x is 3, the statement is true, but when x is 5, the statement is false.
7. “ $5(x + 2) = 5x + 10$ ”, however, is a statement with the truth value of “true.” In this case, no matter what number x is, the statement is true.

8. “She doesn’t have red hair.” is a declarative sentence, but it cannot be classified as true or false because, in this case, we don’t know who “she” is. If “she” were Lucille Ball, the statement would be false. If “she” were Shania Twain, the statement would be true.
9. “The city has a population of over 500,000.” is a declarative sentence, but it cannot be classified as true or false because, in this case, we don’t know what city is being considered.
10. “Ten thousand people in China ate rice on August 25, 2004.” is a statement and its truth value is exactly one of “true” and “false” even though we don’t know which one it is.

Exercise 1.

Classify each of the following 15 sentences as a statement or a non-statement. If it’s a statement, classify it as true or false.

1. Who’s your friend?
2. The Amazon is the longest river in Africa.
3. Mercury is the planet closest to the Sun.
4. She can’t wait to graduate.
5. $x < x + 1$.
6. A platypus is an animal that lives in the jungles of South and Central America.
7. Don’t forget to renew your driver’s license.
8. Shakespeare wrote Hamlet.
9. He didn’t do his homework assignment.
10. No spider has 10 legs.
11. Shut the door.
12. Halley’s Comet returns to the solar system every two years.
13. $2x - 18 = 20$.
14. Abraham Lincoln was president of the United States during the Civil War.
15. Did you go home for the holidays?

1.2 DENIALS

When we want to refer to statements symbolically, we usually use letters such as p , q , or r . If p is a statement, then the **denial** of p is a statement, denoted by $\neg p$ and read “not p ,” with the property that whenever p is true, $\neg p$ is false, and whenever p is false, $\neg p$ is true.

EXAMPLE 1: p : My calculus book has 500 pages.
 $\neg p$: My calculus book does not have 500 pages.

EXAMPLE 2: p : My calculus book has 500 pages.
 q : My calculus book has less than 300 pages.

Why is statement q not $\neg p$?

1.3 TRUTH TABLES

The table below illustrates all of the possible truth values of a statement and its denial and is called a truth table.

p	$\neg p$
T	F
F	T

Table 1.1: Truth Table

1.4 COMPOUND STATEMENTS JOINED BY AND/OR

If p and q are two statements, their **conjunction** is the statement p and q , denoted symbolically by $p \wedge q$. For the conjunction of two statements to be true, both of the statements must be true; otherwise the conjunction is false.

If p and q are two statements, their **disjunction** is the statement p or q , denoted symbolically by $p \vee q$. For the disjunction of two statements to be false, both of the statements must be false; otherwise the disjunction is true. In other words, $p \vee q$ is true when at least one of p and q is true.

The table below illustrates all the possible combinations of truth values for p and q and the corresponding truth values for $p \wedge q$ and for $p \vee q$.

p	q	$p \wedge q$	$p \vee q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Table 1.2: Truth Table for a Conjunction and a Disjunction

EXAMPLE:

Let p be “Memorial Day is the last day of June.”

Let q be “Independence Day is celebrated in July.”

Then $p \wedge q$ is “Memorial Day is the last day of June, _____ Independence Day is celebrated in July.”

The truth value for this statement is _____ because _____.

Also $p \vee q$ is “Memorial Day is the last day of June, _____ Independence Day is celebrated in July.”

The truth value for this statement is _____ because _____.

1.5 LOGICALLY EQUIVALENT STATEMENTS, TAUTOLOGIES, & CONTRADICTIONS

Two statements p and q are **logically equivalent** provided that they have exactly the same truth values in all possible cases; i.e., their truth table columns are identical. Symbolically, for the statement that p and q are logically equivalent, we write $p \iff q$.

A statement is a **tautology** provided that its truth value in all possible cases is True; i.e., in a truth table a tautology would have a column of all T's.

A **contradiction** is a statement which has False truth values in all possible cases; i.e., in a truth table a contradiction would have a column of all F's.

In table 1.3, which statements, if any, are equivalent, and how do you know?

Which, if any, are tautologies, and how do you know?

Which, if any, are contradictions, and how do you know?

p	q	$\neg p$	$\neg q$	$p \vee \neg p$	$p \wedge \neg p$	$p \vee q$	$\neg(p \vee q)$	$\neg p \vee \neg q$	$\neg p \wedge \neg q$
T	T								
T	F								
F	T								
F	F								

Table 1.3: Truth Table Showing a Tautology, a Contradiction, and Equivalent Statements

Since the statement $p \wedge \neg p$ is a contradiction, we say that the statement $\neg p$ contradicts the statement p (or that the statement p contradicts the statement $\neg p$).

By Table 1.3, we see that the statements $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

The statement $\neg(p \vee q)$ is read “It is not the case that p or q ,” while the statement $\neg p \wedge \neg q$ is read “Not p , and not q .”

The statement “It is not the case that p or q ” is a denial of the statement “ p or q ,” but it is **not** a very **useful** statement. The statement “Not p , and not q ” is a much more useful statement, so since the two statements are logically equivalent, when we want to write a **useful** denial of “ p or q ,” we will write “Not p , and not q .”

Note that anytime we have two logically equivalent statements, p and q , we can use q instead of p , and likewise we can use p instead of q .

1.6 DENIALS OF COMPOUND STATEMENTS

Not only is “Not p **and** not q ” the useful denial of “ p **or** q ,” but now we also see by Table 1.4 that “Not p **or** not q ” is the useful denial of “ p **and** q .” The statements that $\neg(p \vee q) \iff \neg p \wedge \neg q$ and that $\neg(p \wedge q) \iff \neg p \vee \neg q$ are known as DeMorgan’s Laws.

EXAMPLES:

Statement 1: My math book has a blue cover, and it contains more than 100 pages.

Denial: _____

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \wedge \neg q$	$\neg p \vee \neg q$
T	T						
T	F						
F	T						
F	F						

Table 1.4: Proof of the Other DeMorgan's Law

Statement 2: Either Joe went to the beach last Sunday, or he went to an Astro's baseball game.

Denial: _____

Statement 3: Either Timmy is 5 and Peggy is 9, or Jimmy is 2 and Maggie is 7.

Denial: _____

Most Useful Denial: _____

Exercise 2.

For each of the following pairs of statements, form the conjunction, write its denial, form the disjunction, and write its denial, **labeling each statement as to what it is**. Then classify the resulting 4 statements as true or false.

1. Venus is the planet closest to the Sun; Mercury is the planet farthest from the Sun.
2. Pete owns a Ford F150 (assume true); Mary owns a Mercedes (assume false).
3. Shakespeare wrote A Tale of Two Cities; Tolstoy wrote War and Peace.
4. Orion is not a sign of the Zodiac; Capricorn is a sign of the Zodiac.

1.7 CONDITIONAL STATEMENTS

The statement “If p , then q .” is known as a **conditional** statement. The statement p is called the **hypothesis**, and the statement q is called the **conclusion**. The symbol for a conditional statement is $p \rightarrow q$, which may also be read as “ p implies q .”

A conditional statement can be thought of as being similar to a promise. If I win the lottery next Wednesday, then I will buy you a stick of gum. The only time I would break my promise is when next Wednesday rolls around, I find that I have won the lottery **and I don’t** buy you a stick of gum. If, when Wednesday rolls around, I find that I didn’t win the lottery, then I can either buy you a stick of gum or not as I choose, but in either case I won’t have broken my promise. In fact, in the case that I don’t win the lottery, I feel that I have not made a promise at all.

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Table 1.5: Truth Table for a Conditional Statement

Table 1.5 on the preceding page shows that the only time that the conditional statement $p \rightarrow q$ is false is when the hypothesis p is true and the conclusion is false. The table also shows that $p \rightarrow q \iff \neg p \vee q$.

EXAMPLE:

“If a polygon is a triangle, then the sum of the measures of its angles is 180° .” is a conditional statement. The hypothesis is “a polygon is a triangle,” and the conclusion is “the sum of the measures of its angles is 180° .” This conditional statement is true because when the hypothesis is true, then the conclusion is also true. A logically equivalent statement to the given conditional statement is “either a polygon is not a triangle or the sum of its angles is 180° .” In mathematics, most axioms and theorems are stated in the “if, then” form rather than in the “either, or” form.

1.7.1 DENIALS OF CONDITIONAL STATEMENTS

By the table below we see that the useful denial of $p \rightarrow q$ is $p \wedge \neg q$. Since $p \rightarrow q$ is logically equivalent to $\neg p \vee q$, its useful denial is the same as for

$$\neg p \vee q.$$

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$	$\neg q$	$p \wedge \neg q$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	F	T	F

Table 1.6: Truth Table for the Denial of a Conditional Statement

EXAMPLES:

Statement 1: If Toronto is the capital of New York, then Boston is the capital of Texas.

Denial: _____

Statement 2: If the Great Salt Lake is in Utah, then Lake Okeechobee is in Oklahoma.

Denial: _____

Exercise 3.

Write the denial of each of the following conditional statements. What is the truth value of each statement and its denial (if you know it)?

1. If Sam Houston was a president of the Republic of Texas, then he was also a governor of the state of Texas.
2. If Manhattan was built in one day, then the Sears Tower in Chicago is the tallest building in the world.
3. If the Sun is a star, then Mars is a star also.
4. If the Amazon River is in Africa, then the Nile River is in South America.
5. If “broccoli” is the name of a vegetable, then “cabbage” is the name of a fruit.

1.7.2 UNIVERSAL AND EXISTENTIAL QUANTIFIERS

“If the city of Austin is the capital of Texas, then it has a population of more than 100,000” and “If a city is the capital of a state, then it has a population of more than 100,000” are both conditional statements, but there is a

difference between them. In the first conditional statement, the declarative sentence “the city of Austin is the capital of Texas” is a statement because it can be classified as true. However, in the second conditional statement, the declarative sentence “a city is the capital of a state” is not a statement because it can not be classified as true or false unless we know which city is being considered. So conditional statements like the second one fall into a different category than what the first one does. We will refer to the category the second conditional falls into as “quantified statements.”

In general, when we want to assert that every element of some set has a certain property, we let the **set of all things to be considered** be called a universal set, denoted by $\mathcal{U}$. In the case of a quantified conditional statement, we are going to consider only those things which satisfy the hypothesis (i. e., make the hypothesis true) since anything which makes the hypothesis false renders the conditional true, regardless of the truth value of the conclusion. So in the case of the second conditional above, our $\mathcal{U}$ is the set consisting of the 50 state capitals.

Now to represent an element of $\mathcal{U}$, we use a letter, called a variable. The letter most often used for a variable is x . The property that the element x is to have is specified by the conclusion of the conditional statement and is denoted by $P(x)$. Then we can also say “for every x in $\mathcal{U}$, $P(x)$.” In symbols, that will be written $(\forall x)(P(x))$.

So the statement “If a city is the capital of a state, then it has a population of more than 100,000” is equivalent in meaning to the statement “For every city x that is the capital of a state, x has a population of more than 100,000.” It is also equivalent to the statement “Every state capital has a population of more than 100,000. The latter two statements are still considered conditional statements even though they are not in the “If, then” form. For each of these statements, x represents an element of the set $\mathcal{U}$ containing the 50 state capitals, and $P(x)$ is “ x has a population of more than 100,000.”

Now the set of all elements x in $\mathcal{U}$ for which $P(x)$ is a true statement is called the solution set S for P . The solution set S may be $\mathcal{U}$, it may be a proper subset of $\mathcal{U}$, or it may be empty.

For example, let’s suppose there is a box of chocolates on the desk, $\mathcal{U}$ represents the set of all chocolates in that box, the variable x represents a piece of candy in $\mathcal{U}$, and $P(x)$ represents the sentence “ x has a caramel center.”

In the case that the solution set S is $\mathcal{U}$, we can say, “If x is a piece of choco-

late in the box, then x has a caramel center,” or we can say, “For every piece x of chocolate in the box, x has a caramel center,” or we can say “Every piece of chocolate in the box has a caramel center.” All 3 statements mean the same thing, and all are true about the chocolates in the box. In this case where S is $\mathcal{U}$, the statement $(\forall x)(P(x))$ is true.

In the case that the solution set S is a proper subset of $\mathcal{U}$, we can say, “For some piece x of chocolate in the box, x has a caramel center,” or we can say “There is a piece x of chocolate in the box such that x has a caramel center,” or we can say, “Some piece of chocolate in the box has a caramel center.” Again all 3 of these statements mean the exactly same thing, and all are true about the chocolates in the box. In this case where S is a proper subset of $\mathcal{U}$, the general statement is denoted by $(\exists x)(P(x))$, which is read “There exists an x such that $P(x)$,” and that statement is true for this case.

In the case that the solution set S is empty, we can say, “There is no piece x of chocolate in the box which has a caramel center,” or we can simply say “No piece of chocolate in the box has a caramel center.” If we change our property $P(x)$, which is “ x has a caramel center,” to $\neg P(x)$, which is “ x does not have a caramel center,” then the solution set S for $\neg P$ would be $\mathcal{U}$. In that case we can say, “For every piece x of chocolate in the box, x does not have a caramel center,” or we can say, “If x is a piece of chocolate in the box, then x does not have a caramel center,” or we can say, “Every piece of chocolate in the box does not have a caramel center.” All of these statements are true in this case and mean the same thing. In this situation where the solution set for $\neg P$ is $\mathcal{U}$, the general statement is denoted by $(\forall x)(\neg P(x))$, which is read “For every x , not $P(x)$.”

Euler diagrams show another way of looking at quantified conditional statements. We still retain the same universal set, but now define a set $\mathcal{V}$ to consist of all those things which make the conclusion of the conditional statement true. So for the candy box, $\mathcal{U}$ is the set of all the chocolates in the box, and $\mathcal{V}$ is the set of all things with a caramel center. Figure 1.1 shows how those two sets could be related.

In the diagram on the left, the statement “every piece of chocolate in the box has a caramel center” is true. Also true is the the statement “some piece of chocolate in the box has a caramel center.” In the diagram in the middle, the statement “some piece of chocolate (A) in the box has a caramel center” is true. Also true is the statement “some piece of chocolate (B) in the box does not have a caramel center.” In the diagram on the right, the statement “no piece of chocolate in the box has a caramel center” is true, and so is the equivalent statement “every piece of chocolate in the box does not have a

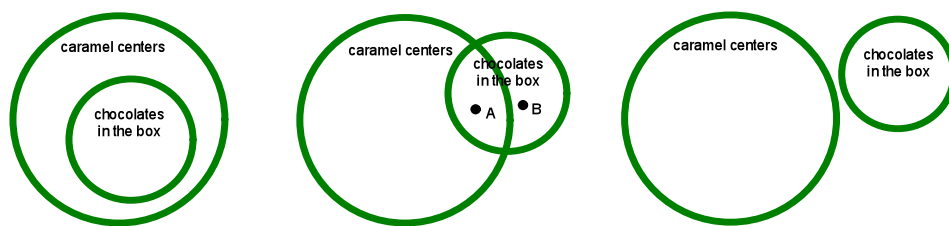


Figure 1.1: Euler Diagrams.

caramel center.”

Words like “**every**,” “**all**,” “**any**,” and “**no**” are called **universal quantifiers**, and the word “**some**,” which means “one or more,” is called an **existential quantifier**. The word “**a**” is NOT considered to be a quantifier of either kind; it is not specific enough. In the statement “a spider has 8 legs,” one usually considers the “a” to mean “every.” However, in the statement “a spider is crawling up the wall behind you,” the “a” (hopefully) only means “some.”

1.7.3 DENIALS OF STATEMENTS WITH UNIVERSAL AND EXISTENTIAL QUANTIFIERS

In order for the statement $(\forall x)(P(x))$ to be false, the solution set S can not be $\mathcal{U}$. Hence there must be at least one element in $\mathcal{U}$ that is not in S ; that is, there is an element x in $\mathcal{U}$ for which $P(x)$ is false and $\neg P(x)$ is true. **A denial of the statement $(\forall x)(P(x))$ is therefore the statement $(\exists x)(\neg P(x))$.** For example, the statement “There is a piece x of chocolate in the box such that x does not have a caramel center.” is a denial of the statement “For every piece x of chocolate in the box, x has a caramel center” and of the statement “If x is a piece of chocolate in the box, then x has a caramel center.” For a denial of the statement “Every piece of chocolate in the box has a caramel center,” we could use the same statement but would more likely use the statement “Some piece of chocolate in the box does not have a caramel center.” It also follows that **a denial of the statement $(\exists x)(\neg P(x))$ is the statement $(\forall x)(P(x))$.**

Likewise, for the statement $(\forall x)(\neg P(x))$ to be false, the solution set S can not be $\mathcal{U}$. Hence there must be at least one element in $\mathcal{U}$ that is not in S ; that is, there is an element x in $\mathcal{U}$ for which $\neg P(x)$ is false and $P(x)$ is true. **A denial of the statement $(\forall x)(\neg P(x))$ is the statement $(\exists x)(P(x))$.** For example, the denial of the statement “For every piece x of chocolate in the

box, x does not have a caramel center” is the statement “There is a piece x of chocolate in the box such that x has a caramel center.” It is also correct to write “For some piece x of chocolate in the box, x does not have a caramel center.” For a denial of the statement “Every piece of chocolate in the box does not have a caramel center,” we would use the statement “Some piece of chocolate in the box has a caramel center.”

For the statement $(\exists x)(P(x))$ to be true, the solution set S can be either $\mathcal{U}$ or a nonempty subset of $\mathcal{U}$. So $(\exists x)(P(x))$ is false when the solution set S is empty. Hence **a denial of the statement $(\exists x)(P(x))$ is the statement $(\forall x)(\neg P(x))$** . Mathematically, this form is preferred over a statement such as “No x in $\mathcal{U}$ has property $P(x)$,” or “There is no x in $\mathcal{U}$ for which property $P(x)$ is true.” So mathematically speaking, we prefer the denial of “Some piece of chocolate in this box has a caramel center.” to be given as “Every piece of chocolate in this box does not have a caramel center.” rather than “No piece of chocolate in this box has a caramel center,” or “There is no piece of chocolate in the box which has a caramel center.” Basically, we want the negative as close to the end of the sentence as possible, and in many cases, the negative can be avoided altogether.

EXAMPLES:

1. All (normal) spiders have 8 legs.

Denial: _____

2. Some book contains at least 1000 pages.

Denial: _____

3. For every polygon P , P has exactly 4 sides.

Denial: _____

4. There is a positive number x such that $\frac{1}{x} > 1$.

Denial: _____

5. If f is a function which is continuous at $x = 5$, then f is differentiable at $x = 5$.

Denial: _____

Exercise 4.

Write the useful denials of each of the following statements.

1. All caterpillars metamorphose into butterflies.
2. There is a mammal which lays eggs.
3. x is an element of set A, or x is an element of set B. (Assume that we know what “ x ” is.)
4. If Boston is the capital of Massachusetts, then Chicago is the capital of Illinois.
5. For every positive integer x , $x^2 \geq 1$.
6. Some flowers are red, and some trees have flowers.
7. If a quadrilateral is a parallelogram, then it has two congruent sides.
8. Either some bird eats insects, or all birds eat grain.
9. There is an atlas which contains 100 maps or more.
10. If a book is a mathematics book, then it contains at least one equation.
11. Seattle is in Indiana, or Salt Lake City is in Utah.
12. Either 61349 is a prime number, or one of its factors is 31.
13. If x is a mammal that swims in the ocean and weighs less than 200 pounds, then x is a seal.
14. Some fish is not less than six inches long.
15. For every real number x , there is a positive integer y such that $x + y > 100$.
16. Some mountain does not have any snow on it at any time of the year.
17. Either a student likes mathematics, or he does not do well in mathematics courses.
18. There are integers x and y such that if n is any positive integer, then either $x - n < y$ or $y - n < x$.
19. For every positive real number x , there is a positive integer n such that $nx > 1$.
20. If x and y are real numbers such that $y - x > 1$, then there is an integer n such that $x < n$ and $n < y$.
21. If ε is a positive number, there is a positive number δ such that for every real number x such that $|x - 2| < \delta$, then $|x^2 - 4| < \varepsilon$.

1.7.4 RELATED CONDITIONAL STATEMENTS

There are three other conditional statements related to the conditional “If p , then q .” These are the **converse** - “If q , then p ,” the **inverse** - “If not p , then not q ,” and the **contrapositive** - “If not q , then not p .” In writing the converse of a given conditional statement, you don’t just switch the hypothesis and the conclusion word for word. In the hypothesis of any conditional statement, the “object or objects under discussion” need to be identified. In the conclusion, that “object” or those “objects” may then be referred to by a pronoun. The resulting statement needs to be grammatically correct.

EXAMPLE

Conditional: If a person is a Texan, then he is an American.

Converse: If a person is an American, then he is a Texan.

Inverse: If a person is not a Texan, then he is not an American.

Contrapositive: If a person is not an American, then he is not a Texan.

Note that in each of the statements above, the object mentioned is a “person,” and this is stated in the hypothesis in every case. Then in the conclusion the person under discussion is referred to by the generic pronoun “he.”

Table 1.7 shows that the given conditional and its contrapositive have identical truth values and are therefore equivalent, while the converse and the inverse also have identical truth values and are likewise equivalent.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

Table 1.7: Truth Table for the Related Conditionals

EXAMPLE:

Conditional: If two integers are both even, then their sum is even.

Converse: _____.

Inverse: _____.

Contrapositive: _____.

In this example the “object or objects under discussion” are “two integers” or “the sum of two integers,” and the appropriate pronoun is either “they” or “their.”

Exercise 5.

For the following conditional statements, write the converse, the inverse, and the contrapositive. Give each statement’s truth value. Which of the 4 statements (the given conditional, the converse, the inverse, and the contrapositive) always have the same truth values?

1. If Newton’s birthday was February 10, then 8731 is a prime number.
2. If the diagonals of a quadrilateral are perpendicular, then the quadrilateral is a square.
3. If two angles of one triangle are congruent to two angles of another triangle, then the two triangles are similar.
4. If 2 lines in space are parallel, then they do not intersect.
5. If an integer is divisible by 6, then it is an even integer.

1.7.5 BICONDITIONAL STATEMENTS

A **biconditional** is a statement of the form “ p if and only if q ,” symbolically denoted by $p \leftrightarrow q$, and is true precisely when p and q have the same truth values.

The phrase “if and only if” is often abbreviated as “iff.”

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

Table 1.8: Truth Table for a Biconditional

The truth table in Table 1.8 shows that the biconditional $p \leftrightarrow q$ is logically equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$.

EXAMPLES:

1. A polygon is a quadrilateral if and only if it has four sides.
2. A heavenly body is a planet if and only if it revolves around the Sun, has enough mass to be nearly round, and has cleared the neighborhood around its orbit.

Definitions are given as biconditional statements, as in the examples above. So when the biconditional is true, we note, by the table, that if p is true, then q is true, and if q is true, then p is true. Likewise, when the biconditional is true, if p is false, then q is false, and if q is false, then p is false.

Chapter 2

METHODS OF PROOF

2.1 PRELIMINARIES

In every mathematical system, there two types of terms: defined and undefined. Examples of undefined terms are set, element, in, number, point, between, and line. Then there are terms which are defined in terms of the undefined terms, such as subset, even integer, ray, angle.

Also, there are three types of statements:

1. **definitions**, which are always of the type of statement $p \leftrightarrow q$,
2. **axioms**, or postulates, which are statements that are simply assumed to be true.
3. **theorems**, which are “proved,” that is, shown to follow logically from axioms, definitions, or other theorems which have already been proved.

Any set of axioms (or postulates) is kept to a minimum. Why do you think that is? Theorems are usually conditional statements, such as $p \rightarrow r$. The hypothesis p contains what is known as the “given information,” that is, the information we have to start out with.

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Table 2.1: Making a Deduction

As you see in Table 2.1, $(p \wedge (p \rightarrow q)) \rightarrow q$ is a tautology. Therefore, if we know that p is true and that $p \rightarrow q$, then we can logically assert that q must be true. This kind of reasoning is known as deductive reasoning.

2.2 DIRECT PROOFS

The following is a template for a direct proof.

Theorem: $p \rightarrow r$

Proof: Let p .	(reason: given information) This is the way every proof starts out.
Then q .	(reason: from some axiom, definition, or previously proved theorem saying $p \rightarrow q$)
Then m .	(reason: from one of the same sources saying that $q \rightarrow m$, or $p \rightarrow m$, or $p \wedge q \rightarrow m$)
$\vdots$	$\vdots$
Thus r .	(similar type of reason) This is the conclusion of the theorem and the way every proof ends.

We now give an example of a direct proof.

Definition E. The integer n is **even** if and only if there is an integer k such that $n = 2 \cdot k$.

Theorem A. If m and n are both even integers, then $m + n$ is an even integer.

Proof:

Let m and n both be even integers.	(given)
Then there is an integer k such that $m = 2k$.	(_____)
and _____	(definition of even integer)
So then $m + n =$ _____	(equal #'s added to equal #'s)
and $m + n = 2 \cdot$ (_____)	(distributive property)
But _____ is an integer,	(_____)
so $m + n$ is an even integer.	(_____)

Therefore, if m and n are even integers, then $m + n$ is even. (This statement is a recap of what we just proved.)

A strategy that is often used to help one understand what a definition or theorem is saying is to find examples to **illustrate** that definition or that

theorem. Find 3 examples of even integers, and show why they are (using the definition, of course). Find 3 examples of integers which are not even. Illustrate the theorem with 3 examples. How many pairs of even integers are there? Could 3 examples prove the theorem is true?

2.3 PROOF BY THE CONTRAPOSITIVE

A proof by the contrapositive is actually a **direct proof** also, just not of the given statement. Instead it is a proof of the equivalent conditional statement called its contrapositive.

Theorem: $p \rightarrow r$ (The equivalent statement is $\neg r \rightarrow \neg p$).

Proof: Let $\neg r$. (reason: given information)
 Then q . (reason)
 Then m . (reason)
 $\vdots$ $\vdots$
 Thus $\neg p$. (reason)
 Therefore, since $\neg r \rightarrow \neg p$, $p \rightarrow r$.

2.4 INDIRECT PROOFS

For the theorem $p \rightarrow r$, the conclusion r is either true or false. Since we want to show that r is true, then in an **indirect proof** we suppose instead that r is false or, what is equivalent, that $\neg r$ is true.

Theorem: $p \rightarrow r$

Proof: Let p . (reason: given information)
 Suppose $\neg r$. (supposition)
 Then q . (reason)
 Then m . (reason)
 $\vdots$ $\vdots$
 Then $w \wedge \neg w$. (reason)
 This is a contradiction. Therefore, the supposition is false, hence r (is true).
 (OR: $\neg w$ contradicts w , so therefore the supposition is false. Hence r .)

In this type of proof, one doesn't usually know beforehand exactly what statement is going to be contradicted. But there **will be** a contradiction, and it will be to either the given information or to some definition, axiom, or

previously proved theorem.

We now give an example of an indirect proof.

Definition O. The integer n is **odd** if and only if there is an integer k such that $n = 2 \cdot k + 1$.

Assume also that we know that an integer is either even or odd, but not both.

Theorem B. If m and n are integers such that $m \cdot n$ is an even integer, then one of m and n is an even integer.

Proof:

Let m and n be integers such that $m \cdot n$ is even.

(given)

Suppose it is not true that one of m and n is an even integer.

(Supposition: $\neg q$)

Then _____.

(useful denial of q
by the assumption
above)

Then _____.

So _____.

So _____.

But this contradicts _____. Therefore, _____, so _____.

Therefore, if $m \cdot n$ is even, then m is even or n is even.

The tautologies in the next exercise will be especially useful when proving the theorems concerning sets in the next chapter.

Exercise 6.

In part A, show that the given statements are tautologies. Then use the tautologies to answer the corresponding questions in part B.

A: 1. $(p \wedge q) \rightarrow p$

2. $(p \wedge q) \rightarrow (p \vee q)$

3. $p \rightarrow (p \vee q)$

4. $p \vee (q \vee r) \leftrightarrow (p \vee q) \vee r$

5. $p \wedge (q \wedge r) \leftrightarrow (p \wedge q) \wedge r$

6. $p \wedge (q \vee r) \leftrightarrow (p \wedge q) \vee (p \wedge r)$

7. $p \vee (q \wedge r) \leftrightarrow (p \vee q) \wedge (p \vee r)$
8. $(p \vee q) \rightarrow r \leftrightarrow (p \rightarrow r) \wedge (q \rightarrow r)$
9. $(\neg r \rightarrow (w \wedge \neg w)) \rightarrow r$ (This is the basis for the indirect proof.)

- B:**
1. If you know “John is here and Judy is here,” then you know “_____.”
 2. If you know “John is here and Judy is here,” then you know “_____.”
 3. If you know “John is here,” then you know “_____.”
 4. “Billy has gone home or Terry is still here, or Joe has not yet arrived.” is logically equivalent to the statement “_____.”
 5. “Billy has gone home, and Terry is still here and Joe has not yet arrived.” is logically equivalent to the statement “_____.”
 6. “Susie is 5 or Josie is 9, and Susie is 5 or Maggie is 7.” is logically equivalent to the statement “_____.”
 7. “Bill is 5 ft. 7 in. tall, and Mary is 4 ft. 10 in. tall or Gordon is 4 ft. 10 in. tall.” is logically equivalent to the statement “_____.”
 8. “If either two integers are both odd or they are both even, then their sum is even.” is logically equivalent to the statement “_____.”

Chapter 3

SETS AND SET OPERATIONS

3.1 SETS, ELEMENTS, AND NOTATION

What we understand a **set** to be is a “collection” of “objects” (called elements of the set) which are characterized by some property that lets us think of the collection as a unit. The property that characterizes the elements must be something that determines precisely whether an object belongs to the set or not. In other words, we say a set must be “well defined.”

We denote sets by capital letters and their elements by lower case letters.

A set may be defined by words, such as $\mathbb{N}$ is the set of all natural numbers. “The set S of all the people in China who ate rice on January 1, 2007” defines a set, but “the set of all good books” does not define a set. Why, or why not?

A set may be defined by a list (the roster method), such as, $A = \{1, 2, 3, 4, 5, 6\}$, or a set may be defined by the set builder notation, such as $B = \{x | x \text{ is a prime number}\}$, read either as “the set of all x such that x is a prime number” or as “the set to which x belongs if and only if x is a prime number.”

In defining a set using the roster method or the set builder notation, braces are always used to denote sets just as parentheses are always used to denote ordered pairs of numbers. We think of the set as being separate from its elements. If a is an object in set A (or an object which belongs to set A), we write “ $a \in A$,” which is read “ a is an element of set A .” If b is not an object in set A (or b doesn’t belong to A), we write “ $b \notin A$,” which is read “ b is not an element of set A .” Also, since the set A is separate from its elements, we write symbolically “ $A \notin A$.”

Now when we define the set S as $S = \{x|P(x)\}$, we know that if $x \in S$, then we know that $P(x)$ is true, and if we know that $P(x)$ is true, then we know that $x \in S$.

3.2 AN AXIOM

Axiom 0. There is a set, denoted by $\emptyset$, called the **empty set** or the null set, which has no elements.

3.3 DEFINITIONS

Definition 1. A is a **subset** of B (denoted by $A \subseteq B$) if and only if every element of A is an element of B .

Definition 2. If A and B are sets, $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Definition 3. If A and B are sets, A is a **proper subset** of B , denoted by $A \subset B$, if and only if $A \subseteq B$, but $A \neq B$.

Definition 4. If A is a finite set, then the number of elements in A is called the **cardinality** of A and is denoted by $|A|$.

Definition 5. If A is a set, then the **power set** of A , denoted by $\mathcal{P}(A)$, is $\{X|X \subseteq A\}$.

Definition 6. If A and B are sets, then the **union** of A and B , denoted by $A \cup B$, is $\{x|x \in A \text{ or } x \in B\}$. So $x \in A \cup B$ if and only if $x \in A$ or $x \in B$.

Definition 7. If A and B are sets, then the **intersection** of A and B , denoted $A \cap B$, is $\{x|x \in A \text{ and } x \in B\}$. So $x \in A \cap B$ if and only if $x \in A$ and $x \in B$.

3.4 VENN DIAGRAMS

In a given problem we may assume that all of the sets we are discussing are subsets of some universal set $\mathcal{U}$. Using a universal set also allows us to draw pictures that represent sets and how they might behave. Such pictures are referred to as Venn Diagrams. Below are Venn diagrams illustrating A

as a subset of B , $A \cup B$, and $A \cap B$ (with $A \cup B$ and $A \cap B$ needing to be shaded in correctly).

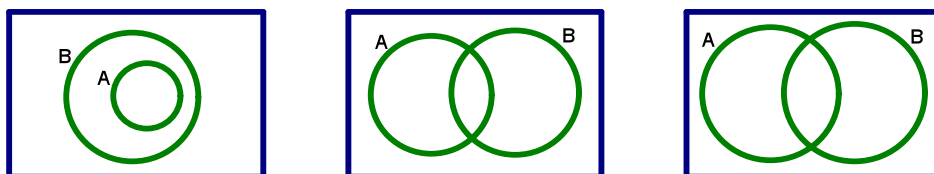


Figure 3.1: $A \subseteq B$, $A \cup B$, and $A \cap B$

3.5 MORE DEFINITIONS WITH VENN DIAGRAMS

Definition 8. If A and B are sets, then A and B are said to be **disjoint** if and only if $A \cap B = \emptyset$.

Definition 9. If A and B are sets, then **the complement of A with respect to B** , denoted by $B \setminus A$, is $\{x | x \in B \text{ but } x \notin A\}$. (Shade in in the figure below.)

Definition 10. If A is a set, then the **complement of A** , denoted A^C , is $\mathcal{U} \setminus A = \{x | x \notin A\}$. (Shade in in the figure below.)

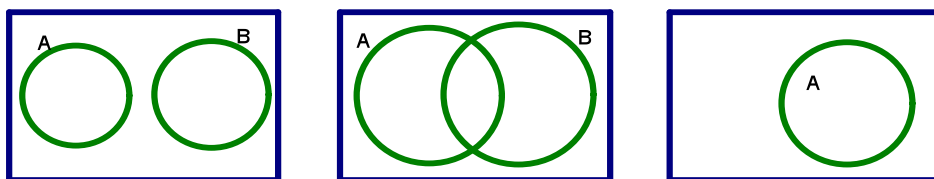


Figure 3.2: $A \cap B = \emptyset$, $B \setminus A$, and A^C

Definition 11. The **symmetric difference** of two sets A and B , denoted by $A \Delta B$, is $\{x | x \in A \text{ but } x \notin B, \text{ or } x \in B \text{ but } x \notin A\}$. (Shade in in the figure below.)

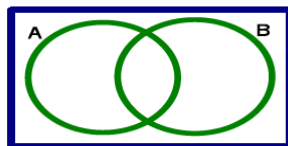
Exercise 7.

1. Let $\mathcal{U} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$, $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{1, 4, 6, 9, 11\}$, and $C = \{1, 5, 9, 12\}$. Find each of the following sets and show your work:

a. A^C

b. $A \cup B$

c. $A \cap B$

Figure 3.3: $A \Delta B$

- | | | |
|---------------------------------|---------------------------------|--------------------------------------|
| d. $B \setminus A$ | e. $A \setminus B$ | f. $(A \cup B)^C$ |
| g. $(A \cap B)^C$ | h. $A^C \cup B^C$ | i. $A^C \cap B^C$ |
| j. $ A $ | k. $A \cup (B \cap C)$ | l. $A \cap (B \cup C)$ |
| m. $(A \cup B) \cap (A \cup C)$ | n. $(A \cap B) \cup (A \cap C)$ | o. $(A \cup B) \setminus (A \cap B)$ |

2. Shade in each of the following sets in a Venn Diagram, and show your work.

- | | | |
|---------------------------------|---------------------------------|------------------------|
| a. $(A \cup B)^C$ | b. $(A \cap B)^C$ | c. $A^C \cup B^C$ |
| d. $A^C \cap B^C$ | e. $A \cup (B \cap C)$ | f. $A \cap (B \cup C)$ |
| g. $(A \cup B) \cap (A \cup C)$ | h. $(A \cap B) \cup (A \cap C)$ | |

Which of the above sets appear to be the same?

3. a. Let $A = \{1, 2, 3\}$. What is $|A|$? _____ List all of the subsets of A . Those are the elements of $\mathcal{P}(A)$. What is $|\mathcal{P}(A)|$? _____
- b. Let $S = \{a, b, c, d\}$. What is $|S|$? _____ List the elements of $\mathcal{P}(S)$. What is $|\mathcal{P}(S)|$? _____
- c. Let B be a set such that $|B| = n$. What is $|\mathcal{P}(B)|$? _____
4. a. Rewrite the statement “every element of A is an element of B ” in the “if p , then q ” form.

_____.

- b. What related conditional statement is equivalent to the statement above? Write that related conditional statement.

_____.

5. The statement in 4 a. above is another way of saying that $A \subseteq B$. So now the definition of $A \not\subseteq B$ is the denial of the statement in 4 a. Write the denial of that statement.

$A \not\subseteq B$ if and only if _____.

6. If A and B are sets, then $x \in A \cup B$ if and only if _____.

so $x \notin A \cup B$ if and only if _____.

7. If A and B are sets, then $x \in A \cap B$ if and only if _____.

so $x \notin A \cap B$ if and only if _____.

8. If A and B are sets, then $A = B$ if and only if _____.

so $A \neq B$ if and only if _____.

9. If A and B are sets, then $x \in B \setminus A$ if and only if _____.

so $x \notin B \setminus A$ if and only if _____.

10. Two sets A and B are disjoint if and only if _____,

so A and B are not disjoint if and only if _____,

3.6 THEOREMS

***Theorem 1.** If A is a set, then

1. $A \subseteq A$
2. $A \cup A = A$
3. $A \cap A = A$
4. $(A^C)^C = A$

***Theorem 2.** If A is a set with no elements, then $A = \emptyset$.

***Theorem 3.** If A is a set, then

1. $\emptyset \subseteq A$
2. $A \cup \emptyset = A$
3. $A \cap \emptyset = \emptyset$

***Theorem 4.** If A is a set, then

1. $A \cup A^C = \mathcal{U}$
2. $A \cap A^C = \emptyset$

***Theorem 5.** If A is a set, then

1. $A \cup \mathcal{U} = \mathcal{U}$
2. $A \cap \mathcal{U} = A$

***Theorem 6.**

1. $\emptyset^C = \mathcal{U}$
2. $\mathcal{U}^C = \emptyset$

***Theorem 7.** If A and B are sets, then

1. $A \cup B = B \cup A$
2. $A \cap B = B \cap A$

***Theorem 8.** If A and B are sets, then

1. $A \cap B \subseteq A$
2. $A \subseteq A \cup B$

Note: The preceding theorem also implies that $A \cap B \subseteq B$ and $B \subseteq A \cup B$.

Theorem 9. If A , B , and C are sets such that $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Theorem 10. If A and B are sets, then $(A \cup B)^C = A^C \cap B^C$.

Theorem 11. If A and B are sets, then $(A \cap B)^C = A^C \cup B^C$.

Theorem 12. If A and B are sets, then $(A \cup B) \setminus B = A \cap B^C$.

Theorem 13. If A, B, C and D are sets such that $A \subseteq B$ and $C \subseteq D$, then $A \cup C \subseteq B \cup D$ and $A \cap C \subseteq B \cap D$.

Theorem 14. If A, B and C are sets, then

1. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
2. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Theorem 15. If A, B and C are sets, then

1. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
2. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Theorem 16. If A and B are sets, then $A \cup B = A$ if and only if $B \subseteq A$.

Theorem 17. If A and B are sets, then $A \subseteq B$ if and only if $B^C \subseteq A^C$.

Theorem 18. If A and B are sets, then $A \subseteq B$ if and only if $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Theorem 19. If A, B and C are sets, then $A \setminus B \subseteq C$ if and only if $A \setminus C \subseteq B$.

3.7 DISPROVING A CONDITIONAL STATEMENT

The statement $(\forall x)(P(x))$ is false when the statement $(\exists x)(\neg P(x))$ is true. One **specific** example of an x which makes $P(x)$ false is called a counterexample, and giving a counterexample is called disproving the statement $(\forall x)(P(x))$. Statements whose truth values are unknown are referred to as **conjectures**.

3.8 CONJECTURES TO PROVE OR DISPROVE

Conjecture 1. If A and B are sets, then $(A \setminus B)^C = B^C \setminus A^C$.

Conjecture 2. If A, B and C are sets such that $A \cup C \subseteq B \cup C$, then $A \subseteq B$.

Conjecture 3. If A, B and C are sets such that $A \cap C \subseteq B \cap C$, then $A \subseteq B$.

Conjecture 4. If A, B and C are sets, then $(A \setminus B) \cup C = A \setminus (B \cup C)$.

Conjecture 5. If A, B and C are sets, then $(A^C \cup B) \cap (B^C \cup C) = A^C \cup C$.

Conjecture 6. If A, B and C are sets, then $(A \cap B^C) \cap C^C = (A \cap C^C) \setminus (B \cap C^C)$.

3.9 VALID AND INVALID ARGUMENTS

What, if anything, is wrong with each of the following conjectures and/or proofs?

Conjecture 7. If A, B , and C are sets such that $A \subseteq B$ and $B \not\subseteq C$, then $A \not\subseteq C$.

“Proof 1:” Let $A = \{1, 3, 5, 7\}$, $B = \{1, 3, 5, 7, 9, 11\}$ and $C = \{5, 7, 9, 11\}$. Then $A \subseteq B$ since every element in A is also in B ; but $B \not\subseteq C$ since 1 is in B , but 1 is not in C . Then we also see that since 1 is in A and not in C , $A \not\subseteq C$.

“Proof 2:” Let A, B and C be sets such that $A \subseteq B$ and $B \not\subseteq C$. Let $x \in A$. Then since $A \subseteq B$, then $x \in B$ also. But since $B \not\subseteq C$, $x \notin C$. Thus $x \in A$, but $x \notin C$. Therefore, $A \not\subseteq C$.

“Proof 3:” As can be seen by the Venn Diagram on the next page, since $A \subseteq B$ and $B \not\subseteq C$, then $A \not\subseteq C$.

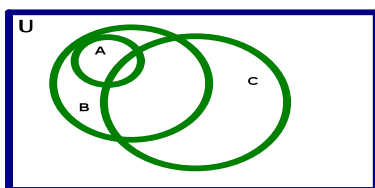


Figure 3.4: “Proof 3.”

Conjecture 8. If A, B , and C are sets, then $A \cup (B \cap C)^C \subseteq (A \cup B^C) \cap (A \cup C^C)$.

“Proof:” Let A, B , and C be sets. Let $x \in A \cup (B \cap C)^C$. Then $x \in A$, or $x \in (B \cap C)^C$. So $x \in A$, or $x \notin B \cap C$. Then $x \in A$, or $x \notin B$ and $x \notin C$. Thus $x \in A$, or $x \in B^C$ and $x \in C^C$. So then $x \in A$ or $x \in B^C$, and $x \in A$ or $x \in C^C$. Thus $x \in A \cup B^C$ and $x \in A \cup C^C$. Hence, $x \in (A \cup B^C) \cap (A \cup C^C)$, and there-

fore, $A \cup (B \cap C)^C \subseteq (A \cup B^C) \cap (A \cup C^C)$.

Conjecture 9. If A, B , and C are sets such that $A \subseteq B$, then $A \cap C^C \subseteq B \cap C^C$.

“Proof:” Let A, B , and C be sets such that $A \subseteq B$. Let $x \in A$. Since $A \subseteq B, x \in B$. Let C be any set that does not contain x . Then $x \notin C$. Then $x \in C^C$. Hence $x \in A$ and $x \in C^C$, so $x \in A \cap C^C$. Also $x \in B$ and $x \in C^C$, so $x \in B \cap C^C$. Consequently, $A \cap C^C \subseteq B \cap C^C$.

Conjecture 10. If A and B are sets, then $\mathcal{P}(A \setminus B) \subseteq \mathcal{P}(A) \setminus \mathcal{P}(B)$.

“Proof:” Let A and B be sets. Let $\{x\} \in \mathcal{P}(A \setminus B)$. Then $\{x\} \subseteq A \setminus B$. So $x \in A \setminus B$. Hence $x \in A$ but $x \notin B$. Then $\{x\} \subseteq A$, but $\{x\} \not\subseteq B$. So $\{x\} \in \mathcal{P}(A)$, but $\{x\} \notin \mathcal{P}(B)$. Thus, $\{x\} \in \mathcal{P}(A) \setminus \mathcal{P}(B)$. Therefore, $\mathcal{P}(A \setminus B) \subseteq \mathcal{P}(A) \setminus \mathcal{P}(B)$.

Chapter 4

RELATIONS AND EQUIVALENCE RELATIONS

4.1 DEFINITIONS DEALING WITH RELATIONS

Definition 12. The **Cartesian product** of two nonempty sets A and B , denoted by $A \times B$, is $\{(a, b) | a \in A \text{ and } b \in B\}$; a is called the **first component** of (a, b) , and b is called the **second component** of (a, b) .

Definition 13. A **relation** $\mathcal{R}$ from A to B is a nonempty subset of $A \times B$. A relation $\mathcal{R}$ which is a subset of $A \times A$ is called a **relation on A** .

Notation. If $(a, b) \in \mathcal{R}$, then we write $a\mathcal{R}b$, and read that as “ a is related to b .”

Definition 14. If $\mathcal{R} \subseteq A \times B$, the **domain** of $\mathcal{R}$ is the set $dom(\mathcal{R}) = \{x \in A | (x, y) \in \mathcal{R} \text{ for some } y \in B\}$ and the **range** of $\mathcal{R}$ is the set $range(\mathcal{R}) = \{y \in B | (x, y) \in \mathcal{R} \text{ for some } x \in A\}$.

Notation.

The **set of all real numbers** is denoted by $\mathbb{R}$, the set $\mathbb{R} \setminus \{0\}$ of non-zero real numbers is denoted by $\mathbb{R}^*$, and the Cartesian product $\mathbb{R} \times \mathbb{R}$ is denoted by $\mathbb{R}^2$. The set of all positive real numbers is denoted by $\mathbb{R}^+$.

The **set of all rational numbers** is denoted by $\mathbb{Q}$, the set $\mathbb{Q} \setminus \{0\}$ of non-zero rational numbers is denoted by $\mathbb{Q}^*$, and the set of all positive rational numbers is denoted by $\mathbb{Q}^+$.

The **set of all integers** is denoted by $\mathbb{Z}$, the set $\mathbb{Z} \setminus \{0\}$ of non-zero integers is denoted by $\mathbb{Z}^*$, and the set of all positive integers is denoted by $\mathbb{Z}^+$.

The **set of all natural numbers** is denoted by $\mathbb{N}$. This is the same set as the set $\mathbb{Z}^+$.

The **set of all complex numbers** is denoted by $\mathbb{C}$, and the set of all non-zero complex numbers is denoted by $\mathbb{C}^*$.

Exercise 8.

Find the domain and range of the given relations.

- | | |
|---|---|
| 1. $\{(1, 2), (3, 4), (5, 6), (7, 8), (9, 10)\}$ | 2. $\{(-1, (p, q)), (-2, (r, s)), (-3, (t, u))\}$ |
| 3. $\mathbb{R} \times \mathbb{Z}$ | 4. $\mathbb{Q} \times \mathbb{N}$ |
| 5. $\{(x, y) \in \mathbb{R}^2 x, y \in [0, 1] \text{ and } x > y\}$ | 6. $\{(x, y) \in \mathbb{R}^2 y = 5x + 2\}$ |
| 7. $\{(x, y) \in \mathbb{R}^2 y = x \}$ | 8. $\{(x, y) \in \mathbb{R}^2 x^2 + y^2 = 1\}$ |
| 9. $\{(x, y) \in \mathbb{R}^2 y \leq \sqrt{x}, x \geq 0\}$ | 10. $\{((a, b), a + b) a, b \in \mathbb{Z}\}$ |

4.2 EQUIVALENCE RELATIONS

Definition 15. A relation $\mathcal{R}$ on a non-empty set A is an **equivalence relation on A** if and only if the following 3 conditions are satisfied:

- | | |
|--|------------------------|
| i. if $a \in A$, then $a\mathcal{R}a$. | (reflexive property) |
| ii. if $a\mathcal{R}b$, then $b\mathcal{R}a$. | (symmetric property) |
| iii. if $a\mathcal{R}b$ and $b\mathcal{R}c$, then $a\mathcal{R}c$. | (transitive property). |

An equivalence relation will be denoted by $\sim$ rather than by $\mathcal{R}$.

Definition 16. If $\sim$ is an equivalence relation on the set A and $a \in A$, then the **equivalence class** of a for $\sim$ is the set $[a] = \{x \in A | x \sim a\}$.

Definition 17. If A is a non-empty set, a **partition** of A is a collection C of non-empty subsets of A such that

- every element of A is an element of some subset of A that belongs to C , and
- if S_1 and S_2 are two elements of C , then $S_1 \cap S_2 = \emptyset$.

EXAMPLE: Either prove that the following relation is an equivalence relation on the given set, or prove that it is not; in either case, test all 3 conditions required for a relation to be an equivalence relation.

Let $a\mathcal{R}b$ on $\mathbb{Z}$ if and only if $a + n = b$ for some integer $n \geq 0$.

Give examples of pairs (a, b) such that a is related to b : $(_, _)$, $(_, _)$, $(_, _)$.

Give examples of pairs (a, b) such that a is not related to b : $(_, _)$, $(_, _)$, $(_, _)$.

i. Reflexive property: Let $a = 3$. Then $a + _ = _$.
 $_$ is an integer greater than or equal to 0. Therefore $3\mathcal{R}3$.
 So $\mathcal{R}$ is reflexive.

ii. Symmetric property: Let $a = 3$ and $b = 10$. Then $3 + 7 = 10$ and 7 is
 an integer greater than or equal to 0, thus $3\mathcal{R}10$.
 But suppose $10\mathcal{R}3$. Then there is a non-negative integer n such
 that $10 + n = 3$. Then $n = _$, but $_$ is an
 integer greater than or equal to 0. Therefore, $10 \not\mathcal{R} 3$, so
 $\mathcal{R}$ is not symmetric.

iii. Transitive property: Let $a = 3$ and $b = 10$.
 Then because $a\mathcal{R}b$, $3 + 7 = 10$,
 and because $b\mathcal{R}c$, $10 + 1 = 11$,
 So $3 + 7 + 1 = 11$.
 11 is an integer greater than or equal to 0.
 Therefore, $3\mathcal{R}11$, so $\mathcal{R}$ is transitive.
 Hence $\mathcal{R}$ is an equivalence relation on $\mathbb{Z}$.

Exercise 9.

- A. In problems 1-14, **be able** to find 3 pairs of elements in the set which **are** related to each other, and show why they are related. **Be able** to find 3 pairs of elements in the set which **are not** related to each other, and show why they are not related.
- B. Either prove that the following relations are equivalence relations on the given sets, or find a counterexample. Test all 3 conditions required for a relation to be an equivalence relation. If the relation is an equivalence relation, can you **describe** some of the equivalence classes?
1. Let $a\mathcal{R}b$ on $\mathbb{Z}$ if and only if there is an integer k such that $b = ak$.
 2. Let $a\mathcal{R}b$ on $\mathbb{Z}$ iff $\gcd(a, b) = 1$. (Note: $\gcd(a, b)$ stands for the greatest common divisor of a and b .)
 3. Let $a\mathcal{R}b$ on $\mathbb{Z}$ iff $a = b(2^k)$ for some $k \in \mathbb{Z}$.
 4. Let $a\mathcal{R}b$ on $\mathbb{Z}$ iff $a = 5m - b$ for some integer m .
 5. Let $a\mathcal{R}b$ on $\mathbb{Q}^+$ iff $a = b^k$ for some $k \in \mathbb{Q}$.
 6. Let $a\mathcal{R}b$ on $\mathbb{R}$ iff $a - 1 \leq b \leq a + 1$.
 7. Let $a\mathcal{R}b$ on $\mathbb{R}$ iff $a = kb$ for some $k \in \mathbb{Q}^*$.

Prove that the following relations are equivalence relations on the given sets, and **describe** some of the equivalence classes. In the case that there are only finitely many such classes, describe all of them.

8. Let $a \sim b$ on $\mathbb{Q}$ iff $a - b = m$ for some integer m .
9. Let $a \sim b$ on $\mathbb{Z}$ iff $|a| = |b|$.
10. Let $a \sim b$ on $\mathbb{Z}$ iff $(a - b) = 7k$ for some integer k . (This relation is known as congruence modulo 7, and instead of writing $a \sim b$, we would write $a \equiv b \pmod{7}$).
11. Let $(a, b) \sim (c, d)$ on $\mathbb{N}^2$ iff $a + d = b + c$.
12. Let $(a, b) \sim (c, d)$ on $\mathbb{Z} \times \mathbb{Z}^*$ iff $ad = bc$.
13. Let $(a, b) \sim (c, d)$ on $\mathbb{R}^2$ iff $a^2 + b = c^2 + d$.
14. Let $(a, b, c) \sim (d, e, f)$ on $\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ iff $2a + 7b - 3c = 2d + 7e - 3f$.

4.3 THEOREMS ABOUT EQUIVALENCE RELATIONS

Theorem 20. If $\sim$ is an equivalence relation on a set A and $a \sim b$, then $[a] = [b]$.

Theorem 21. If $\sim$ is an equivalence relation on a set A , then the collection of all equivalence classes of A for $\sim$ is a partition of A , and if P is a partition of A , then there is an equivalence relation on A for which P is the collection of all its equivalence classes.

Chapter 5

MAPPINGS AND BIJECTIONS

5.1 DEFINITIONS, NOTES, AND COMMENTS

Definition 18. If A and B are two non-empty sets, a **mapping** f (function, map) from A to B is a subset of $A \times B$ such that if $a \in A$, there is exactly one ordered pair in f having a as its first component. If A and B are sets of numbers, then we usually use the word “function” instead of the word “mapping.”

Notation. “The mapping f from set A to set B ” and “ f is a mapping from set A to set B ” are both denoted by “ $f : A \rightarrow B$.”

Definition 19. If (a, b) is an ordered pair in f , then b is denoted by $f(a)$ and is called the **image of a under f** while a is called a **pre-image** of b .

Definition 20. The set A is called the **domain** of f , and the set B is called the **co-domain** of f . The **range** of f (or **image of A under f**) is the set $f(A) = \{b \in B \mid b = f(a) \text{ for some } a \text{ in } A\}$.

Definition 21. If $C \subseteq A$, then the **image of C under f** is the set $f(C) = \{b \in B \mid b = f(a) \text{ for some } a \text{ in } C\}$, while if $D \subseteq B$, the **pre-image of D** is the set $f^{-1}(D) = \{a \in A \mid f(a) = d \text{ for some } d \text{ in } D\}$.

Definition 22. If $b \in B$, then $f^{-1}(\{b\})$ is called the **fiber** of f over b .

Definition 23. If $C \subseteq A$ and $f : A \rightarrow B$, then the **restriction** of f to C will be denoted by $f|_C$.

EXAMPLES

- a. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2x^2 - 5x - 1$ for all x in $\mathbb{R}$. Then

what is

- i. the image of 5?
- ii. a pre-image of 5?
- iii. the fiber of f over 5?
- iv. the range of f ?
- v. $f(C)$ where $C = (-1, 1)$?
- vi. $f^{-1}([-1, 1])$?

b. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $g(x) = 7 - 3x - x^2$ for all x in $\mathbb{R}$. Then what is

- i. the image of -4 ?
- ii. a pre-image of -4 ?
- iii. the fiber of g over -4 ?
- iv. the range of g ?
- v. $g(C)$ where $C = (-3, 2)$?
- vi. $g^{-1}([-5, 7])$?

c. Let $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $h(x, y) = x + y$ for all (x, y) in $\mathbb{R}^2$. Then what is

- i. the image of $(1, 2)$?
- ii. a pre-image of 7?
- iii. the fiber of h over -4 ?
- iv. the range of h ?
- v. $h(C)$ where $C = \{(x, y) | x^2 + y^2 = 1\}$?
- vi. $h^{-1}([-\sqrt{2}, \sqrt{2}])$?

Definition 24. If $f : A \rightarrow B$ and $g : A \rightarrow B$, then $f = g$ if and only if for all $a \in A$, $f(a) = g(a)$.

Definition 25. If $f : A \rightarrow B$ and $g : B \rightarrow C$, then the **composite** map $g \circ f : A \rightarrow C$ is defined by $g \circ f(a) = g(f(a))$ for all $a \in A$.

Definition 26. A function $f : A \rightarrow B$ is **one-to-one** (1-to-1, 1-1, injective, an injection) means that if x_1 and x_2 are elements of A such that $f(x_1) = f(x_2)$, then $x_1 = x_2$ (or, by the contrapositive, if x_1 and x_2 are elements of A such

that $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$).

Definition 27. A function $f : A \rightarrow B$ is **onto** (surjective, a surjection) if and only if every element in B is the image of some element in A (i.e., $f(A) = B$).

Definition 28. A function $f : A \rightarrow B$ is a **bijection** if and only if f is both one-to-one and onto. (In this case, f is sometimes referred to as a one-to-one correspondence.)

Definition 29. Two sets A and B have the **same cardinality** if and only if there is a bijection from A to B . Sets which have the same cardinality are often referred to as being **equally numerous**.

5.2 DENIALS

1. A function $f : A \rightarrow B$ is **not** one-to-one means that _____
_____.
2. A function $f : A \rightarrow B$ is **not** onto if and only if _____
_____.

EXAMPLES: Determine if the given function is one-to-one, onto, both, or neither, and prove your answers are correct.

1. Let $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ where $\alpha(x) = 5 - 2x - 2x^2$ for all x in $\mathbb{R}$.

1 - 1: Let $\alpha(x) = 5$. Then _____
so $x =$ _____. Then $\alpha(0) = 5 - 0 - 0 = 5$, and $\alpha(\text{_____}) =$
_____, so $\alpha(0) = \alpha(\text{_____})$, but 0 _____.
Therefore α _____ one-to-one.

Onto: _____ is an element of $\mathbb{R}$. Suppose α is onto. Then there
is _____ such
that _____. Then _____

_____, so $x =$ _____ or $x =$ _____.
But _____.

Therefore α _____ onto.

2. Let $\beta : \mathbb{R} \rightarrow \mathbb{R}$ where $\beta(x) = \frac{5-2x}{3}$ for all x in $\mathbb{R}$.

1 - 1: Let $\beta(x_1) = \beta(x_2)$. Then _____
 so _____,
 and _____, then _____.
 Therefore β _____ one-to-one.

Onto: Let _____. Suppose β is onto. Then there is _____ such that
 _____. Then _____
 _____, so $x =$ _____.
 Now _____ is in $\mathbb{R}$ because _____,
 and $\beta(\text{_____}) =$ _____.
 Therefore β _____ onto.

Exercise 10.

For each of the mappings given in problems 1) through 23), be able to show the following: 3 things in the domain of the mapping, 3 things not in the domain of the mapping, and 3 ordered pairs that belong to the mapping. Then determine if the mapping is one-to-one, onto, both, or neither.

1. $\psi : \mathbb{Z} \rightarrow \mathbb{Z}$ where $\psi(x) = 2x - 1$ for all x in $\mathbb{Z}$.
2. $\omega : \mathbb{Q} \rightarrow \mathbb{Q}$ where $\omega(x) = 3x + 5$ for all x in $\mathbb{Q}$.
3. $\tau : \mathbb{R} \rightarrow \mathbb{R}$ where $\tau(x) = \frac{3x-5}{4}$ for all x in $\mathbb{R}$.
4. $\phi : \mathbb{Q} \rightarrow \mathbb{Q}$ where $\phi(x) = \frac{8x}{3} + \frac{5}{2}$ for all x in $\mathbb{Q}$.
5. $\pi : \mathbb{R}^* \rightarrow \mathbb{R}$ where $\pi(x) = \frac{x-3}{x}$ for all x in $\mathbb{R}^*$.
6. $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ where $\alpha(x) = x^2 - 2x - 1$ for all x in $\mathbb{R}$.
7. $\omega : \mathbb{R} \rightarrow \mathbb{R}^+ \cup \{0\}$ where $\omega(x) = x^2 - 4x + 4$ for all x in $\mathbb{R}$.
8. $\alpha : [-2, \infty) \rightarrow [-7, \infty)$ where $\alpha(x) = x^2 + 4x - 3$ for all x in $[-2, \infty)$.
9. $\vartheta : \mathbb{R} \setminus \{3\} \rightarrow \mathbb{R}$ where $\vartheta(x) = \frac{2x+1}{3-x}$ for all x in $\mathbb{R} \setminus \{3\}$.

10. $\eta : \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R} \setminus \{2\}$ where $\eta(x) = \frac{2x}{x-2}$ for all x in $\mathbb{R} \setminus \{2\}$.
11. $\nu : \mathbb{R} \setminus \{\frac{-7}{2}\} \rightarrow \mathbb{R} \setminus \{\frac{3}{2}\}$ where $\nu(x) = \frac{3x-5}{2x+7}$ for all x in $\mathbb{R} \setminus \{\frac{-7}{2}\}$.
12. $\sigma : \mathbb{R} \setminus \{-2\} \rightarrow \mathbb{R}^*$ where $\sigma(x) = \frac{5}{x+2}$ for all x in $\mathbb{R} \setminus \{-2\}$.
13. $\gamma : \mathbb{Z} \times \mathbb{Z}^* \rightarrow \mathbb{Q}$ where $\gamma(m, n) = \frac{m}{n}$ for all (m, n) in $\mathbb{Z} \times \mathbb{Z}^*$.
14. $\lambda : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ where $\lambda(m, n) = 3m + 2n$ for all (m, n) in $\mathbb{Z} \times \mathbb{Z}$.
15. $\mu : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ where $\mu(m, n) = 4m + 2n$ for all (m, n) in $\mathbb{Z} \times \mathbb{Z}$.
16. $\delta : \mathbb{R} \rightarrow \mathbb{R}^2$ where $\delta(x) = (x+1, x-1)$ for all x in $\mathbb{R}$.
17. $\sigma : \mathbb{R}^2 \rightarrow \mathbb{R}$ where $\sigma(x, y) = \frac{x+y}{2}$ for all (x, y) in $\mathbb{R}^2$.
18. $\kappa : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ where $\kappa(m, n) = (2m, n-m)$ for all (m, n) in $\mathbb{Z} \times \mathbb{Z}$.
19. $\beta : \mathbb{R} \rightarrow \mathbb{R}$ where, for all x in $\mathbb{R}$, $\beta(x) = \begin{cases} 2-x & \text{if } x \leq 1 \\ \frac{1}{x} & \text{if } x > 1 \end{cases}$.
20. $\lambda : \mathbb{R} \rightarrow \mathbb{R}$ where, for all x in $\mathbb{R}$, $\lambda(x) = \begin{cases} \frac{6x+1}{3x-3} & \text{if } x \neq 1 \\ 2 & \text{if } x = 1 \end{cases}$.
21. $\eta : \mathbb{Z} \rightarrow \mathbb{Z}$, where for all n in $\mathbb{Z}$, $\eta(n) = \begin{cases} n+1 & \text{if } n \text{ is even} \\ n-1 & \text{if } n \text{ is odd} \end{cases}$.
22. $\phi : \mathbb{R} \rightarrow \mathbb{R}$, where for all x in $\mathbb{R}$, $\phi(x) = \begin{cases} 2x & \text{if } x \text{ is rational} \\ x & \text{if } x \text{ is irrational} \end{cases}$.
23. $\psi : \mathbb{R} \rightarrow \mathbb{R}$, where for all x in $\mathbb{R}$, $\psi(x) = \begin{cases} 2x & \text{if } x \text{ is an integer} \\ 3x & \text{otherwise} \end{cases}$.
24. Let A and B be non-empty sets and $f : A \rightarrow B$ be an onto map. Prove that the relation $\sim$ defined by $a \sim b$ iff $f(a) = f(b)$ is an equivalence relation on A . Can you describe some of the equivalence classes?
25. Let $\mathcal{F}(\mathbb{R}) = \{f | f : \mathbb{R} \rightarrow \mathbb{R}\}$ and let $f \sim g$ in $\mathcal{F}(\mathbb{R})$ if and only if $f(0) = g(0)$. Prove that the relation $\sim$ just defined is an equivalence relation on $\mathcal{F}(\mathbb{R})$. Can you describe some of the equivalence classes?

5.3 A THEOREM ABOUT FUNCTIONS

Theorem 22. Let $f : A \rightarrow B$, $g : B \rightarrow C$ and $h : C \rightarrow D$. Then

1. $(h \circ g) \circ f = h \circ (g \circ f)$,
2. if f and g are one-to-one, then $g \circ f$ is one-to-one,
3. if f and g are onto, then $g \circ f$ is onto,
4. if f and g are bijections, then $g \circ f$ is a bijection, and
5. if f is a bijection and $f^{-1} : B \rightarrow A$ defined by $f^{-1}(b) = a$ where (a, b) is an element of f , then f^{-1} is also a bijection.

5.4 BINARY OPERATIONS

Definition 30. A *binary operation* $*$ on a set S is a mapping from $S \times S$ to S . Instead of writing $*(s, t) = u$, we write $s * t = u$.

Definition 31. A binary operation $*$ on a set S is *commutative* if and only if $s * t = t * s$ for every s and t in S .

Definition 32. A binary operation $*$ on a set S is *associative* if and only if $(s * t) * u = s * (t * u)$ for every s, t and u in S .

Exercise 11.

Do the following definitions for $*$ yield binary operations on the given sets?

1. on $\mathbb{N}$, $a * b = a - b$.
2. on $\mathbb{Q}$, $a * b = a/b$.
3. on $\mathbb{R}^+$, $a * b = ab$.
4. on $\mathbb{R}^+$, $a * b = a \ln b$.

Are the following binary operations commutative or associative, both, or neither? For each operation, select 2 elements in the set and carry out the operation.

5. on $\mathbb{Z}$, $a * b = a - b$.
6. on $\mathbb{R}$, $a * b = a + b + ab$.
7. on $\mathbb{Q}$, $a * b = \frac{a+b}{2}$.
8. on $\mathbb{N}^2$, $(a, b) * (c, d) = (ac + bd, ad + bc)$.
9. on $\mathbb{Z} \times \mathbb{Z}^*$, $(a, b) * (c, d) = (ad + bc, bd)$.
10. on $\mathbb{R}^* \times \mathbb{R}$, $(a, b) * (c, d) = (ac, bc + d)$.
11. on $\mathbb{R}^3$, $(a, b, c) * (d, e, f) = (bf - ce, cd - af, ae - bd)$.
12. on $G = \{\emptyset, A, B, \mathcal{U}\}$, where $A = \{a\}$, $B = \{b\}$, $\mathcal{U} = \{a, b\}$, and $C * D = C \cup D$.
13. on $G = \{\emptyset, A, B, \mathcal{U}\}$, where $A = \{a\}$, $B = \{b\}$, $\mathcal{U} = \{a, b\}$, and $C * D = C \cap D^C$.
14. on $M(2, \mathbb{Z})$ of 2×2 matrices with integer entries and $A * B = A + B$ (i.e., matrix addition).
15. on $M(2, \mathbb{Z})$ of 2×2 matrices with integer entries and $A * B = A \cdot B$ (i.e., matrix multiplication).
16. on $\mathcal{B}$, $f * g = f \circ g$, where $\mathcal{B}$ denotes the collection of all bijections from S to S , and $S = \{1, 2, 3, 4, 5\}$.
17. on S , the set of equivalence classes modulo 7, $[a] * [b] = [a + b]$. (See exercise 9, problem 10.)

Chapter 6

THE SYSTEM $\mathbb{R}$ OF REAL NUMBERS

6.1 AXIOMS AND DEFINITIONS

Axiom R-I. Let $\mathbb{R}$ be a set whose elements will be called real numbers. The relation $=$ is an equivalence relation on $\mathbb{R}$, and there are 2 binary operations, $+$ and $\cdot$, on $\mathbb{R}$ such that

1. If each of a and b is a real number, then $a + b$ is a real number and ab is a real number.
2. $a + b = b + a$ and $ab = ba$ for all real numbers a and b .
3. $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$ for all real numbers a, b and c .
4. There is a real number 0 such that $a + 0 = 0 + a = a$ for all real numbers a .
5. If a is a real number, there is a real number $-a$, known as the opposite of a , such that $a + (-a) = (-a) + a = 0$.
6. $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$ for all real numbers a, b , and c .
7. There is a real number 1 , different from 0 , such that $1 \cdot a = a \cdot 1 = a$ for every real number a .
8. If a is a real number not 0 , there is a real number $\frac{1}{a}$, known as the reciprocal of a , such that $a \cdot \frac{1}{a} = \frac{1}{a} \cdot a = 1$.

Note that if given the equation $x = b$, with the reflexive property of equality we can say $x + a = x + a$, and then using the principle of substitution, we can get $x + a = b + a$. But instead of doing both, we will go directly from $x = b$ to $x + a = b + a$, and just list substitution as the reason, but at

the same time not forgetting that the reflexive property of $=$ was also needed.

Likewise we will go from $x = b$ directly to $ax = ab$, and just give the reason as being substitution.

6.2 THEOREMS

Theorem 1. If a is a real number and x is a real number such that $a + x = a$ or $x + a = a$, then $x = 0$.

Theorem 2. If a is a real number and x is a real number such that $a + x = 0$ or $x + a = 0$, then $x = -a$.

Theorem 3. If a is a real number, then $-(-a) = a$.

Theorem 4. $-0 = 0$

Theorem 5. If each of a , b and c is a real number such that $a + b = a + c$, then $b = c$.

Theorem 6. If each of a and b is real number, there is a one and only one real number x such that $x + a = b$.

Theorem 7. If a is a real number not 0 and x is a real number such that $ax = a$, then $x = 1$.

Theorem 8. If a is a real number not 0 and x is a real number such that $ax = 1$, then $x = \frac{1}{a}$.

Theorem 9. If each of a , b and c is a real number such that $a \neq 0$ and $ab = ac$, then $b = c$.

Theorem 10. If a is any real number, then $a \cdot 0 = 0 \cdot a = 0$.

Theorem 11. If each of a and b is real number such that $ab = 0$, then $a = 0$ or $b = 0$.

Theorem 12. If each of a and b is a real number such that $a \neq 0$, then there

is a one and only one real number x such that $ax = b$.

Theorem 13. If a is a real number, then $(^{-}1)a = ^{-}a$.

Corollary 14. If each of a and b is a real number, then

- 1) $^{-}(a+b) = ^{-}a + ^{-}b$.
- 2) $(^{-}a)b = a(^{-}b) = ^{-}(ab)$.
- 3) $(^{-}a)(^{-}b) = ab$.

Definition 1. If a is a not zero, we define $\frac{b}{a} = b \cdot \frac{1}{a} = \frac{1}{a} \cdot b$. In particular, then $\frac{a}{a} = a \cdot \frac{1}{a} = 1$.

Lemma 15. If each of b and d is a real number such that $b \neq 0$ and $d \neq 0$, then $\frac{1}{b} \cdot \frac{1}{d} = \frac{1}{bd}$.

Theorem 16. If each of a, b, c and d is a real number such that $b \neq 0$ and $d \neq 0$, then $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$ and $\frac{a}{b} + \frac{c}{d} = \frac{ad + cb}{bd}$.

Theorem 17. If each of a and b is a real number such that $b \neq 0$, then $^{-}\left(\frac{a}{b}\right) = \frac{^{-}a}{b} = \frac{a}{^{-}b}$, and $\frac{^{-}a}{^{-}b} = \frac{a}{b}$.

Theorem 18. If each of a, b , and c is a real number such that $a \neq 0$ and $c \neq 0$, then $\frac{ab}{ac} = \frac{b}{c}$.

Definition 2. We define $a - b = a + ^{-}b$.

6.3 THE ORDER AXIOM AND MORE DEFINITIONS

Axiom R-II. There is a subset P of the set $\mathbb{R}$ of real numbers such that

1. if each of a and b is an element of P , then $a + b$ is an element of P and ab is an element of P , and

2. if x is a real number, one and only one of the following three statements is true:
- a. x is in P .
 - b. $\neg x$ is in P .
 - c. $x = 0$.

Definition 3. If each of a and b is a real number, then $a < b$ means that there is an element p of P such that $a + p = b$.

Definition 4. Also $b > a$ means $a < b$.

6.4 ORDER THEOREMS

Theorem 19. $\neg 1$ is not in P , and 1 is in P ,

Prove Theorems 20 through 25 using the definition of “ $<$ ”.

Theorem 20. If each of a and b is a real number, then just one of the following is true: $a = b$, $a < b$, or $b < a$.

Theorem 21. If a is a real number, then a is in P if and only if $a > 0$. In particular, $1 > 0$.

Theorem 22. If a , b , and c are real numbers such that $a < b$ and $b < c$, then $a < c$.

Theorem 23. If a , b , and c are real numbers such that $a < b$, then $a + c < b + c$.

Theorem 24. If a , b , and c are real numbers such that $a < b$ and $c > 0$, then $ac < bc$.

Theorem 25. If a , b , and c are real numbers such that $a < b$ and $c < 0$, then $bc < ac$. (or $ac > bc$.)

Now after Theorem 25, do not use the definition of “ $<$ ” anymore. Instead use Theorems 19 through 25 as needed, assuming that you know that the

hypothesis of the theorem being used is true in the theorem being proved. Also, after this point, it won't be necessary to show every step you make using parts of Axiom I or Theorems 1 - 18. Just use the term "simplifying" as the reason instead of specifying which parts of Axiom I or which of those theorems you actually used.

Theorem 26. If a , b , c , and d are real numbers such that $a < b$ and $c < d$, then $a + c < b + d$.

Theorem 27. If a and b are real numbers such that $a < b$, then $-b < -a$; if $a < 0$, then $-a > 0$; and if $a > 0$, then $-a < 0$. In particular, then $-1 < 0$.

Corollary 28.1. If x and y are real numbers such that $x > 0$ and $y > 0$, or $x < 0$ and $y < 0$, then $xy > 0$.

Corollary 28.2. If x and y are real numbers such that $x > 0$ and $y < 0$, or $x < 0$ and $y > 0$, then $xy < 0$.

Corollary 28.3. If x is a real number not 0, then $xx > 0$ (i.e., $x^2 > 0$).

Theorem 29. If a is a positive number, then $a + \frac{1}{a} \geq 2$.

Theorem 30. If a is a real number, then if $a > 0$, then $\frac{1}{a} > 0$, and if $a < 0$, then $\frac{1}{a} < 0$.

Theorem 31. If x and y are real numbers such that $xy > 0$, then either $x > 0$ and $y > 0$, or $x < 0$ and $y < 0$.

Theorem 32. If x and y are real numbers such that $xy < 0$, then either $x < 0$ and $y > 0$, or $x > 0$ and $y < 0$.

Theorem 33. If a , b , and c are real numbers such that $ac < bc$ and $c > 0$, then $a < b$.

Theorem 34. If a , b , and c are real numbers such that $ac < bc$ and $c < 0$, then $a > b$.

Definition 5. If a and b are real numbers such that $a < b$, then c is **between a and b** if and only if $a < c$ and $c < b$. This is also denoted by $a < c < b$.

Theorem 35. If a and b are real numbers such that $a < b$, then $a < \frac{a+b}{2} < b$.

Theorem 36. If a and b are real numbers such that $0 < a < b$, then $\frac{1}{a} > \frac{1}{b}$.

Theorem 37. If a , b , and c are real numbers such that $a > 0$ and $b \geq 1$, then $c - \frac{a}{b} \geq c - a$.

6.5 THE NATURAL NUMBERS AND THE LEAST NATURAL NUMBER AXIOM

Definition 6. A non-empty set S contains a smallest element if and only if there is an element s in S such that for every element x in S $s \leq x$.

Definition 7. A non-empty set S contains a largest element if and only if there is an element g in S such that for every element x in S $x \leq g$.

Axiom R-III. There is a subset $\mathbb{N}$ of P , known as the set of natural numbers, such that

1. 1 is an element of $\mathbb{N}$.
2. if n is an element of $\mathbb{N}$, then $n + 1$ is an element of $\mathbb{N}$, and if n is greater than 1, then $n - 1$ is also an element of $\mathbb{N}$.
3. if m and n are elements of $\mathbb{N}$, then $m + n \in \mathbb{N}$ and $mn \in \mathbb{N}$.
4. if n is an element of $\mathbb{N}$, then $n \geq 1$.

5. (The Least Natural Number Axiom) every non-empty subset of $\mathbb{N}$ contains a smallest element.

Theorem 38. If n is an element of $\mathbb{N}$, then there is no element k of $\mathbb{N}$ such that $n < k < n + 1$.

Definition 8. The real number a is an **integer** if and only if either a is in $\mathbb{N}$ (thus, being called a positive integer), a is 0, or $-a$ is in $\mathbb{N}$.

Definition 9. If a and b are integers such that $b \neq 0$, b **divides** a if and only if there is an integer k such that $a = bk$.

Definition 10. A positive integer n is **prime** if and only if it has exactly 2 positive divisors.

EXAMPLES:

1 is not a prime because _____.

2 is a prime because _____.

9 is not a prime because _____.

12 is not a prime because _____.

Definition 11. A positive integer n is called **composite** if it has more than 2 positive divisors. Then it has a divisor different from n and 1.

6.6 THE FIRST PRINCIPLE OF MATHEMATICAL INDUCTION

Theorem 39. (The First Principle of Mathematical Induction) If S is a subset of $\mathbb{N}$ such that

- i.) 1 is in S and ii.) if k is in S , then $k + 1$ is in S ,

then $S = \mathbb{N}$.

Exercise 12.

Prove each of the following by the First Principle of Mathematical Induction.

1. If n is a natural number, then $1 + 2 + 3 + 4 + \cdots + n = \frac{n(n+1)}{2}$.
2. If n is a natural number, then $1 + 3 + 5 + 7 + \cdots + (2n-1) = n^2$.
3. If n is a natural number, then $2 + 5 + 8 + 11 + \cdots + (3n-1) = \frac{n(3n+1)}{2}$.
4. If n is a natural number, then $3 + 11 + 19 + \cdots + (8n-5) = 4n^2 - n$.
5. If n is a natural number, then $-1 + 2 - 3 + 4 - \cdots + (-1)^n n = \frac{[(-1)^n(2n+1) - 1]}{4}$.
6. If n is a natural number, then $1 + 3 + 6 + 10 + \cdots + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}$.
7. If n is a natural number, then $1^2 + 2^2 + 3^2 + 4^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$.
8. If n is a natural number, then $1^3 + 2^3 + 3^3 + 4^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$.
9. If n is a natural number, then $1^3 + 3^3 + 5^3 + 7^3 + \cdots + (2n-1)^3 = n^2(2n^2 - 1)$.
10. If n is a natural number, then $1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^{n-1}} = 2 - \frac{1}{2^{n-1}}$.
11. If n is a natural number, then $\frac{1}{1(2)} + \frac{1}{2(3)} + \frac{1}{3(4)} + \frac{1}{4(5)} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$.
12. Let x and y be real numbers such that $0 < x < y$. Then if n is a natural number, then $0 < x^n < y^n$.
13. If n is a natural number, then 3 divides $5^n - 2^n$. (Use the definition of divides given in Section 6.5.)
14. If n is a natural number, then 5 divides $8^n - 3^n$.
15. If n is a natural number, then 7 divides $11^n - 4^n$.
16. If n is a natural number, then 2 divides $3^n - 1$.
17. If n is a natural number and x and y are different integers, then $(x-y)$ divides $x^n - y^n$.

18. Let x and a be real numbers. Then for every natural number n ,

$$x^{n+1} - a^{n+1} = (x - a) \left(\sum_{i=0}^n x^{n-i} a^i \right).$$

(Note: $t \cdot \left(\sum_{i=1}^n f(i) \right) = \sum_{i=1}^n (t \cdot f(i))$ and $\sum_{i=1}^{n+1} f(i) = \sum_{i=1}^n f(i) + f(n+1)$.)

What, if anything, is wrong with the following three conjectures and/or "proofs?"

20. **Conjecture A.** If n is a natural number, then $n^2 + n + 41$ is prime.

"Proof:" Let $n = 1$. Then $n^2 + n + 41 = 1^2 + 1 + 41 = 43$, which is prime.
 Let $n = 2$. Then $n^2 + n + 41 = 2^2 + 2 + 41 = 47$, which is prime.
 Let $n = 3$. Then $n^2 + n + 41 = 3^2 + 3 + 41 = 53$, which is prime.
 Let $n = 4$. Then $n^2 + n + 41 = 4^2 + 4 + 41 = 61$, which is prime.
 Let $n = 5$. Then $n^2 + n + 41 = 5^2 + 5 + 41 = 71$, which is prime.
 Let $n = 6$. Then $n^2 + n + 41 = 6^2 + 6 + 41 = 83$, which is prime.
 Let $n = 7$. Then $n^2 + n + 41 = 7^2 + 7 + 41 = 97$, which is prime.
 Let $n = 20$. Then $n^2 + n + 41 = 20^2 + 20 + 41 = 461$, which is prime.
 Let $n = 35$. Then $n^2 + n + 41 = 35^2 + 35 + 41 = 1301$, which is prime.
 Let $n = 42$. Then $n^2 + n + 41 = 42^2 + 42 + 41 = 1847$, which is prime.
 Let $n = 51$. Then $n^2 + n + 41 = 51^2 + 51 + 41 = 2693$, which is prime.
 Let $n = 100$. Then $n^2 + n + 41 = 100^2 + 100 + 41 = 10141$, which is prime.

Therefore if n is a natural number, then $n^2 + n + 41$ is prime.

21. **Conjecture B.** If n is a natural number, then

$$(2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1) + \cdots + (2 \cdot n + 1) = n^2 + 2n + 1.$$

"Proof:" Let $S = \{n \in \mathbb{N} \mid (2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1) + \cdots + (2 \cdot n + 1) = n^2 + 2n + 1\}$.

Let k be an element of S . Then

$$(2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1) + \cdots + (2 \cdot k + 1) = k^2 + 2k + 1.$$

So then

$$[(2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1) + \cdots + (2 \cdot k + 1)] + (2(k+1) + 1) = [k^2 + 2k + 1] + (2(k+1) + 1).$$

$$\text{So } (2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1) + \cdots + (2(k+1) + 1) = (k+1)^2 + 2(k+1) + 1.$$

Therefore $k+1$ is in S . So $S = \mathbb{N}$, and therefore, for every natural number n , $(2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1) + \cdots + (2 \cdot n + 1) = n^2 + 2n + 1$.

22. **Conjecture C.** All people are the same height.

“Proof:” Let $S = \{n \in \mathbb{N} \mid \text{for every set } A \text{ of } n \text{ people, all the people in set } A \text{ are the same height}\}$.

Let $n = 1$. Let A be a set containing 1 person. Then all the people in set A are the same height.

Now let k be an element of set S . Then for every set of k people, all the people in the set are the same height.

Now let A be a set containing $k + 1$ people. Let p be a person in set A .

Now the set $A \setminus \{p\}$ is a set containing k people.

Since k is in S , all the people in the set $A \setminus \{p\}$ are the same height.

Now let q be a person in set A different from p . The set $A \setminus \{q\}$ is a set of k people and all of the people in this set are the same height.

The set $A \setminus \{p\}$ and the set $A \setminus \{q\}$ have $k - 1$ people in common and they are all the same height as both p and q . So p and q are the same height. Therefore all the people in set A are the same height.

Hence $k + 1$ is in S , and therefore $S = \mathbb{N}$,

Therefore for any n , a set of n people are all the same height.

6.7 THE EXTENDED FIRST PRINCIPLE OF MATHEMATICAL INDUCTION

Theorem 40. Let $m \in \mathbb{N}$. If S is a subset of $\mathbb{N}$ such that

- i.) m is in S , and
- ii.) if $k \geq m$ and $k \in S$, then $k + 1 \in S$,

then $S = \{n \in \mathbb{N} \mid n \geq m\}$.

Exercise 13.

Prove each of the following by the Extended First Principle of Mathematical Induction.

1. If n is a natural number such that $n \geq 6$, then $(n + 1)^2 \leq 2^n$.
2. If n is a natural number such that $n \geq 2$, then $1 + 2^n < 3^n$.
3. If n is a natural number such that $n \geq 3$, then $(1 + \frac{1}{n})^n < n$.
4. If n is a natural number such that $n \geq 5$, then $n = 2k + 3m$ for some whole numbers k and m .

5. If n is a natural number such that $n \geq 4$, then $n! > 2^n$.
6. If x is a real number such that $x > -1$, then if n is a natural number, then $(1+x)^n \geq 1+nx$.
7. If n is a natural number such that $n \geq 2$, then $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$

6.8 THE SECOND PRINCIPLES OF MATHEMATICAL INDUCTION

Theorem 41. If S is a subset of $\mathbb{N}$ such that

- i.) 1 is in S , and
- ii.) if for every k in $\mathbb{N}$ such that $\{1, 2, 3, \dots, k\} \subseteq S$, then $k+1$ is in S

then $S = \mathbb{N}$.

Theorem 42. If m is a natural number and S is a subset of $\mathbb{N}$ such that

- i.) m is in S , and
- ii.) if for every k in $\mathbb{N}$ such that $k \geq m$ and $\{m, m+1, m+2, m+3, \dots, k\} \subseteq S$, then $k+1$ is in S ,

then $S = \{n \in \mathbb{N} | n \geq m\}$.

Theorem 43. (the Fundamental Theorem of Arithmetic) Every natural number greater than 1 is either prime or the product of primes.

6.9 RATIONAL NUMBERS

Definition 12. The real number a is a **rational number** if and only if there are integers p and q with $q \neq 0$ such that $a = \frac{p}{q}$.

Definition 13. If a and b are integers, the **greatest common divisor** of a and b is a positive integer d such that

- i.) d divides a and d divides b , and
- ii.) if c divides a and c divides b , then $c \leq d$.

Theorem 44. If d is the greatest common divisor of a and b , and $a = dk$ and $b = dm$, then the greatest common divisor of k and m is 1.

Definition 14. If m and n are integers, and $\gcd(m, n) = 1$, then $\frac{m}{n}$ is said to be in lowest terms.

Theorem 45. Every rational number can be put in lowest terms.

Definition 15. If x is a real number, $x^1 = x$, and if $n \geq 1$, then $x^{n+1} = x^n \cdot x$.

Definition 16. If x is a positive real number such that $x^2 = n$, then x will be denoted by $\sqrt{n}$.

Theorem 46. There is no rational number x such that $x^2 = 2$.

So now we know that $\sqrt{2}$ is not a rational number. But is there really a number which is $\sqrt{2}$? Well, by the Pythagorean Theorem, we know that a right triangle both of whose legs have length 1 has a hypotenuse with length $\sqrt{2}$, so we do want such a number to exist and be “real”. However, the problem is that the set $\mathbb{Q}$ of rational numbers also satisfies all three of the axioms for real numbers that we have so far. This means that we need another axiom which will help us fill in the “holes” in the real number line. There are a couple of ways to do this, but the one most often used is known as the Completeness Axiom. Another way is to use the Dedekind Cut Axiom. If one assumes one of these two axioms, then the other one can be proved as a theorem.

Consider the following two sequences of numbers.

$$\{a_n\} = \{1, 1.4, 1.41, 1.414, 1.4142, 1.41421, \dots\} \text{ and}$$

$$\{b_n\} = \{2, 1.5, 1.42, 1.415, 1.4143, 1.41422, \dots\}$$

Note that every element in $\{a_n\}$ is smaller than every element in $\{b_n\}$. Also note that $a_n - b_n = \frac{1}{10^{n-1}}$. Is there a rational number between all the elements of $\{a_n\}$ and all the elements of $\{b_n\}$? Any number at all?

6.10 THE COMPLETENESS AXIOM

Definition 17. The real number M is an **upper bound** of a nonempty set S of real numbers provided that if $x \in S$, then $x \leq M$. When a nonempty set S has an upper bound, it is said to be bounded above.

Definition 18. The real number L is a **least upper bound** of a nonempty set S of real numbers if and only if

- i.) L is an upper bound of S and
- ii.) if M is an upper bound of S , then $L \leq M$.

(What is the contrapositive of the statement in Definition 16, part ii?)

Definition 19. The real number M is a **lower bound** of a nonempty set S of real numbers provided that if $x \in S$, then $x \geq M$. When a nonempty set S has a lower bound, it is said to be bounded below.

Definition 20. The real number L is a **greatest lower bound** of a nonempty set S of real numbers if and only if

- i.) L is a lower bound of S , and
- ii.) if M is a lower bound of S , then $M \leq L$.

Definition 21. A nonempty set S of real numbers is said to be bounded when it has both an upper bound and a lower bound.

Axiom R-IV. (The Completeness Axiom) If S is a nonempty set of real numbers which has an upper bound, then S has a least upper bound.

Exercise 14.

Define each of the following.

1. The real number M is NOT an **upper bound** of a nonempty set S of real numbers means that _____.
_____.
2. The real number L is NOT a **least upper bound** of a nonempty set S of real numbers means that _____.

_____.
3. The real number M is NOT a **lower bound** of a nonempty set S of real numbers means that _____.
_____.
4. The real number L is NOT a **greatest lower bound** of a nonempty set S of real numbers means that _____.

_____.
5. A nonempty set S of real numbers is NOT bounded means that _____.
_____.

6.11 THEOREMS

Theorem 47. The set $\mathbb{N}$ of natural numbers has no upper bound.

Exercise 15.

Prove each of the following statements.

1. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{n^2 - 2}{n} \text{ for some natural number } n \right\}$, then S has no upper bound.
2. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{n^2 - 3}{n} \text{ for some natural number } n \right\}$, then S has no upper bound.
3. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{n^2 - 4}{2n} \text{ for some natural number } n \right\}$, then S has no upper bound.
4. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{n^2 - 9}{4n} \text{ for some natural number } n \right\}$, then S has no upper bound.
5. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{n^2 - 10}{3n} \text{ for some natural number } n \right\}$, then S has no upper bound.
6. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{2n^2 - 7}{3n} \text{ for some natural number } n \right\}$, then S has no upper bound.
7. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{3n^2 - 9}{5n} \text{ for some natural number } n \right\}$, then S has no upper bound.
8. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{7n^2 - 1}{3n} \text{ for some natural number } n \right\}$, then S has no upper bound.
9. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{5n^2 - 3}{2n} \text{ for some natural number } n \right\}$, then S has no upper bound.
10. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{5n^2 - 7}{10n} \text{ for some natural number } n \right\}$, then S has no upper bound.

Theorem 48. If $a > 0$, then there is a natural number n such that $an > 1$.

Theorem 49. If $0 < a < b$, then there is a natural number n such that $an > b$.

Theorem 50. If $\varepsilon > 0$, then there is a natural number n such that $\frac{1}{n} < \varepsilon$.

Theorem 51. If each of B and L is a least upper bound of the nonempty set S of real numbers, then $B = L$.

Lemma 52. If B is the least upper bound of the nonempty set S of real numbers and $\varepsilon > 0$, then $B - \varepsilon$ is not an upper bound of S .

Theorem 53. If the real number B is an upper bound of the nonempty set S of real numbers, then B is the least upper bound of S if and only if for every positive number ε there is an x in S such that $x > B - \varepsilon$.

Exercise 16.

1. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{2n-9}{2n} \text{ for some natural number } n \right\}$, then 1 is the least upper bound of S .
2. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{2n}{n+1} \text{ for some natural number } n \right\}$, then 2 is the least upper bound of S .
3. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{n}{2n+1} \text{ for some natural number } n \right\}$, then $\frac{1}{2}$ is the least upper bound of S .
4. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{n-1}{n+8} \text{ for some natural number } n \right\}$, then 1 is the least upper bound of S .
5. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{5n-1}{2n} \text{ for some natural number } n \right\}$, then $\frac{5}{2}$ is the least upper bound of S .
6. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{n}{3n+1} \text{ for some natural number } n \right\}$, then $\frac{1}{3}$ is the least upper bound of S .
7. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{2n}{4n+2} \text{ for some natural number } n \right\}$, then $\frac{1}{2}$ is the least upper bound of S .

8. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{4n}{3n+5} \text{ for some natural number } n \right\}$, then $\frac{4}{3}$ is the least upper bound of S .
9. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{3n-5}{2n+1} \text{ for some natural number } n \right\}$, then $\frac{3}{2}$ is the least upper bound of S .
10. If $S = \left\{ x \in \mathbb{R} \mid x = \frac{7n-4}{n+2} \text{ for some natural number } n \right\}$, then 7 is the least upper bound of S .

Lemma 54. If L is a positive real number such that $L^2 > 2$, then there is a natural number n such that $2 < (L - \frac{1}{n})^2 < L^2$.

Lemma 55. If L is a positive real number such that $L \geq 1$ and $L^2 < 2$, then there is a natural number n such that $L^2 < (L + \frac{1}{n})^2 < 2$.

Theorem 56. There is a real number x such that $x^2 = 2$.

Theorem 57. If $\varepsilon > 0$, then there is a natural number n such that $\frac{\sqrt{2}}{n} < \varepsilon$.

Theorem 58. If $0 < a < b$ and $b - a > 1$, then there is a natural number n such that $a < n < b$.

Theorem 59. If $0 < a < b$ and $\varepsilon > 0$ such that $b - a > \varepsilon$, there is a natural number m such that $a < m\varepsilon < b$.

Theorem 60. If $0 < a < b$, then there is a rational number between a and b .

Theorem 61. If $0 < a < b$, then there is an irrational number between a and b .

Theorem 62. If S is a nonempty set of real numbers which has a lower bound, then S has a greatest lower bound.

Theorem 63. If S_1 and S_2 are two nonempty disjoint sets of real numbers such that

- i.) $S_1 \cup S_2 = \mathbb{R}$,

ii.) if x is an element of S_1 , then $x < y$ for every y in S_2 ,

then there is a number L which is either the largest element in S_1 or the smallest element in S_2 . (This statement is also known as the Dedekind Cut Axiom)

Definition 22. The set I is called a **closed interval** if and only if there are 2 distinct real numbers a and b such that $I = \{x \in \mathbb{R} | x = a, x = b, \text{ or } a < x < b\}$. I is also denoted by $[a, b]$, and a and b are called the endpoints of I .

Theorem 64. If $I_1, I_2, I_3, I_4, \dots$ is an infinite sequence of closed intervals such that if n is a natural number, then $I_{n+1} \subseteq I_n$, then there is a number s which belongs to every interval in the sequence.

APPENDIX A

OTHER FORMS FOR A CONDITIONAL OR BICONDITIONAL STATEMENT

Other ways of saying “If p , then q ” are:

p implies q	when p , then q
q if p	q whenever p
p only if q	p is sufficient for q
q is necessary for p	for q , p is sufficient
for p , q is necessary	

Another way of saying “ p if and only if q ” is “ p is necessary and sufficient for q .”

For example, all of the following statements are equivalent:

If a polygon is a triangle, then it has 3 sides.
A polygon has three sides whenever it is a triangle.
A polygon is a triangle only if it has 3 sides.
A polygon has 3 sides if it is a triangle.
That a polygon is a triangle implies that it has 3 sides.
When a polygon is a triangle, then it has 3 sides.
For a polygon to be a triangle, it is necessary that it have 3 sides.
For a polygon to have 3 sides, it is sufficient that it be a triangle.

Exercise 17.

Write each of the following conditional statements in the “if p , then q ” form; then underline the hypothesis once and the conclusion twice.

1. It always rains whenever we have a vacation.
2. Having a GPA of 2.5 is sufficient for you to enroll in that class.
3. If n is a natural number, then $n \geq 1$.

4. You can borrow books from the library only if you have a library card.
5. When it rains, it pours.
6. It is necessary that you have auto insurance in order to renew your car license.
7. A number is even if it is divisible by 2.
8. That you can reach something on the top shelf implies that you are over 6 feet tall.
9. In order to be able to vote in the national elections, it is sufficient that a U. S. citizen be 18 years old.
10. In order not to pay income tax, you must earn less than \$5000 per year.

Notes to the Instructor

(No specific notes here yet.)